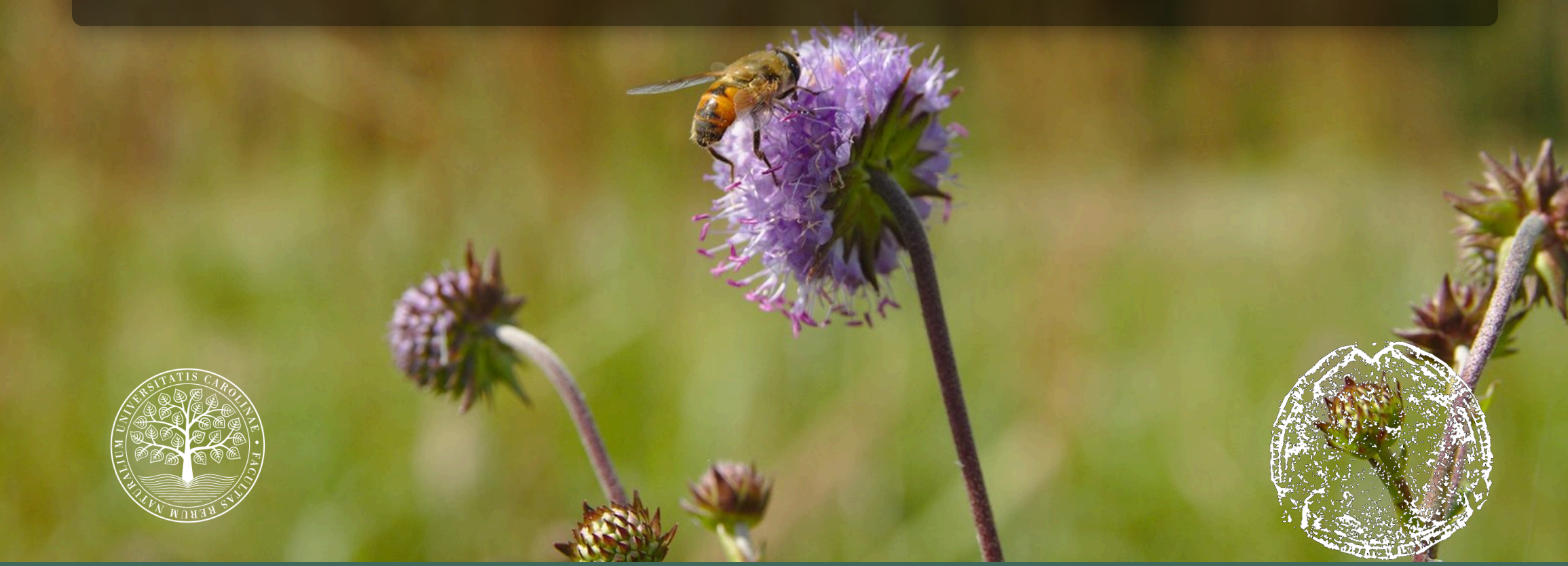
Architecture of stability: 15 years of pollinator sharing reveal a role of stable core species

Jakub Štenc[1,2,3], Sailee P. Sakhalkar[4], Jiří Hadrava[2], Michael Mikát[2], Eva Matoušková[1], Helena Pijálková[2], Tadeáš Ryšan[2,5], Petr Švanda[1,2], Martin Freudenfeld[1], Eva Horčičková[6], Zdeněk Janovský[7], Jakub Straka[2], Jordi Bosch[3]

[1]Charles University (Botany) | [2]Charles University (Zoology) | [3]CREAF, Spain



Interactions

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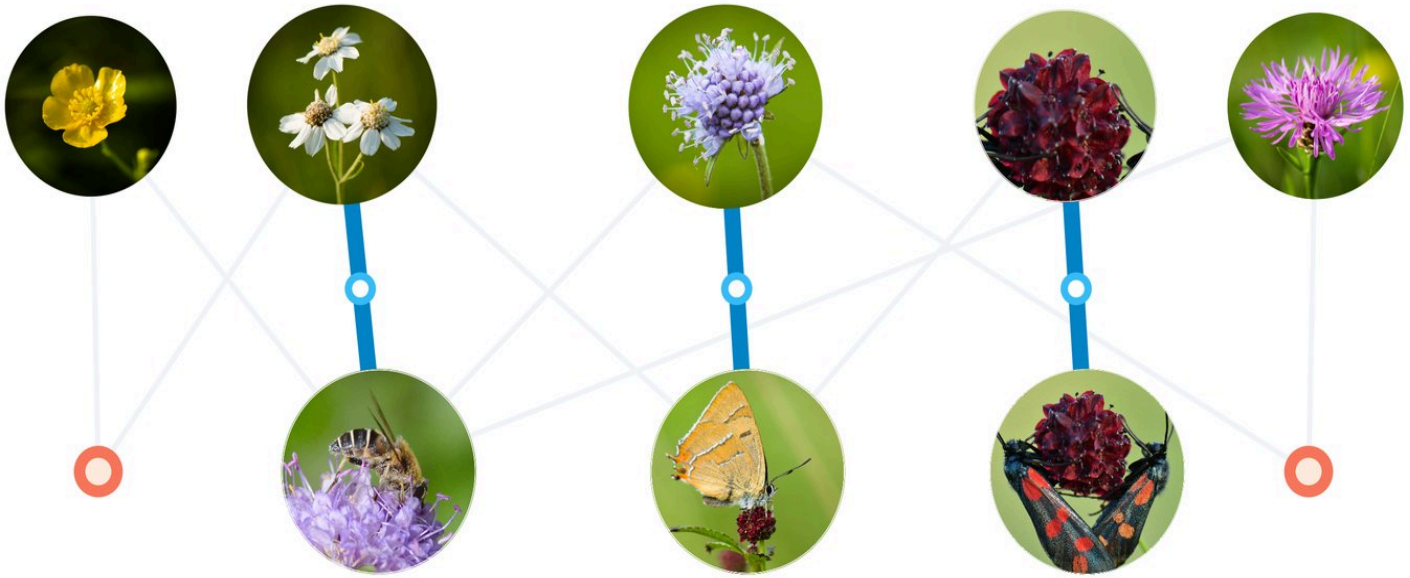

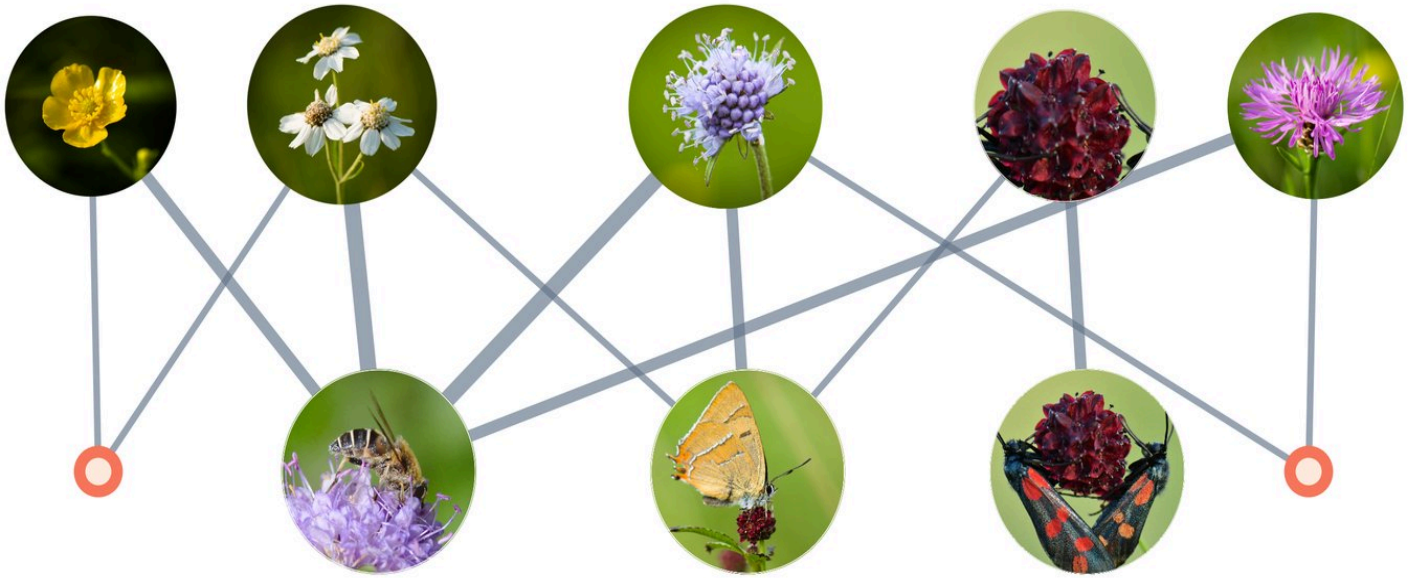

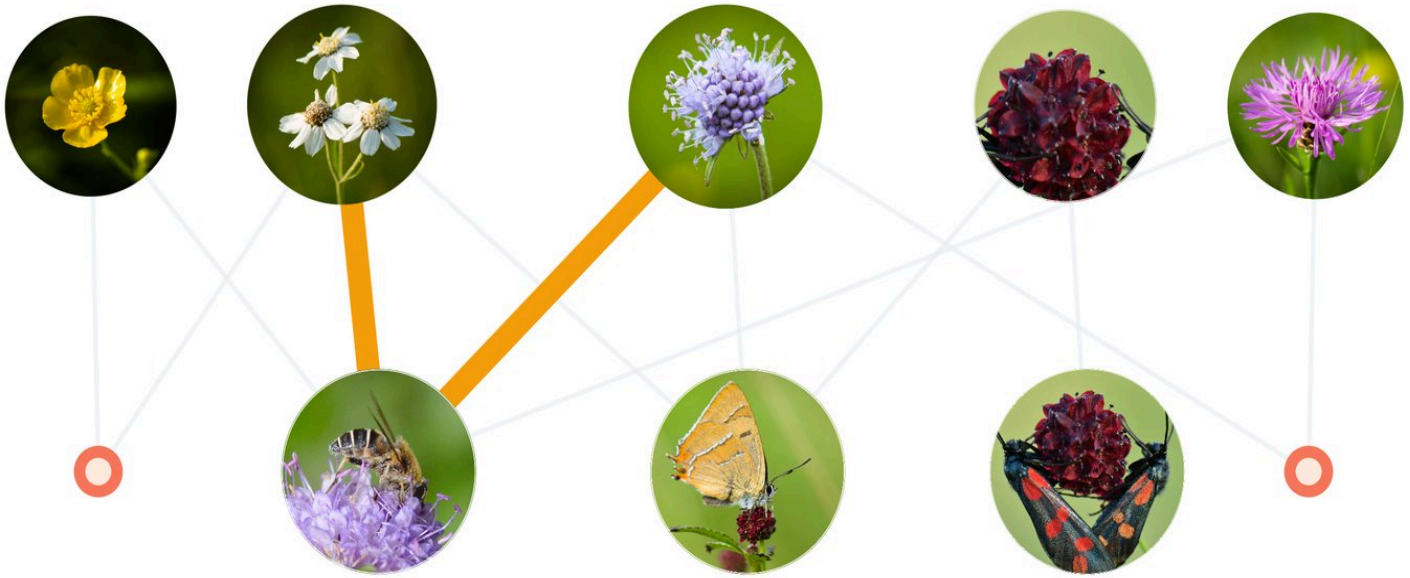

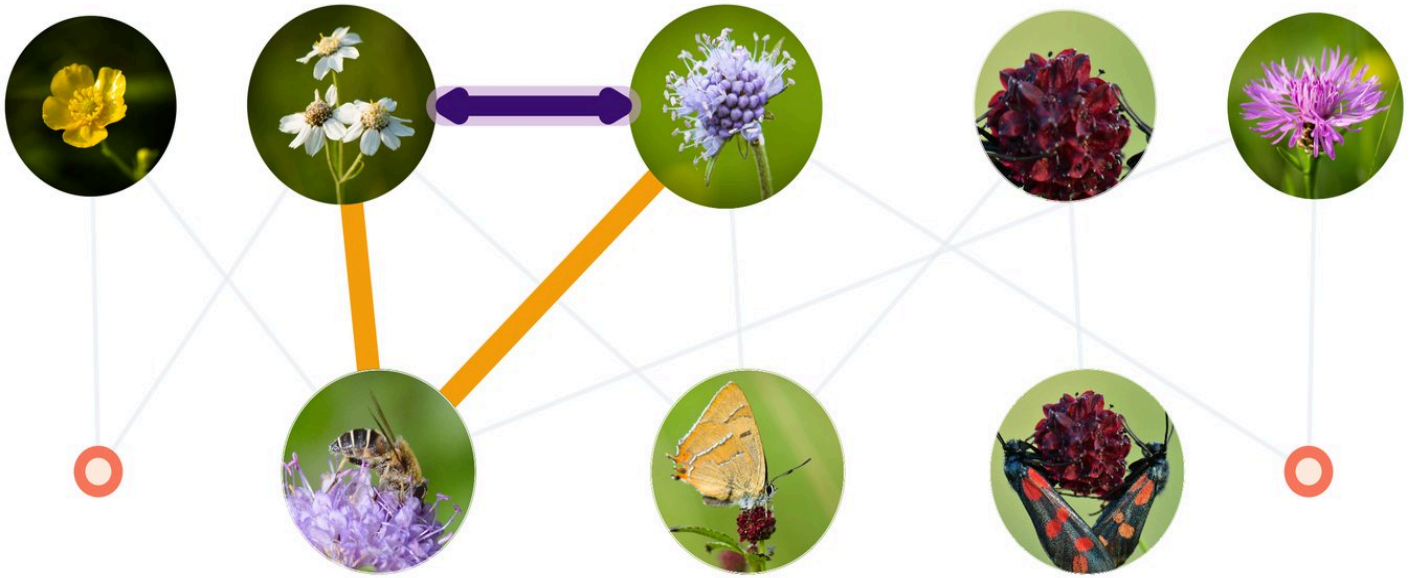

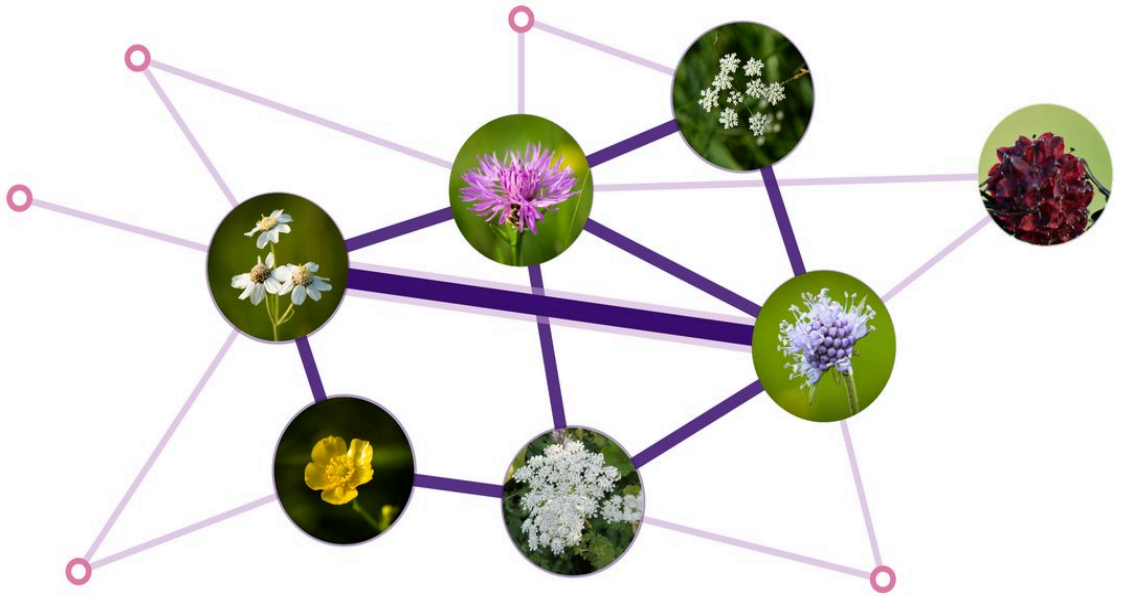

Airports and Meadows

Airports and Meadows

✈️ European Aviation Network

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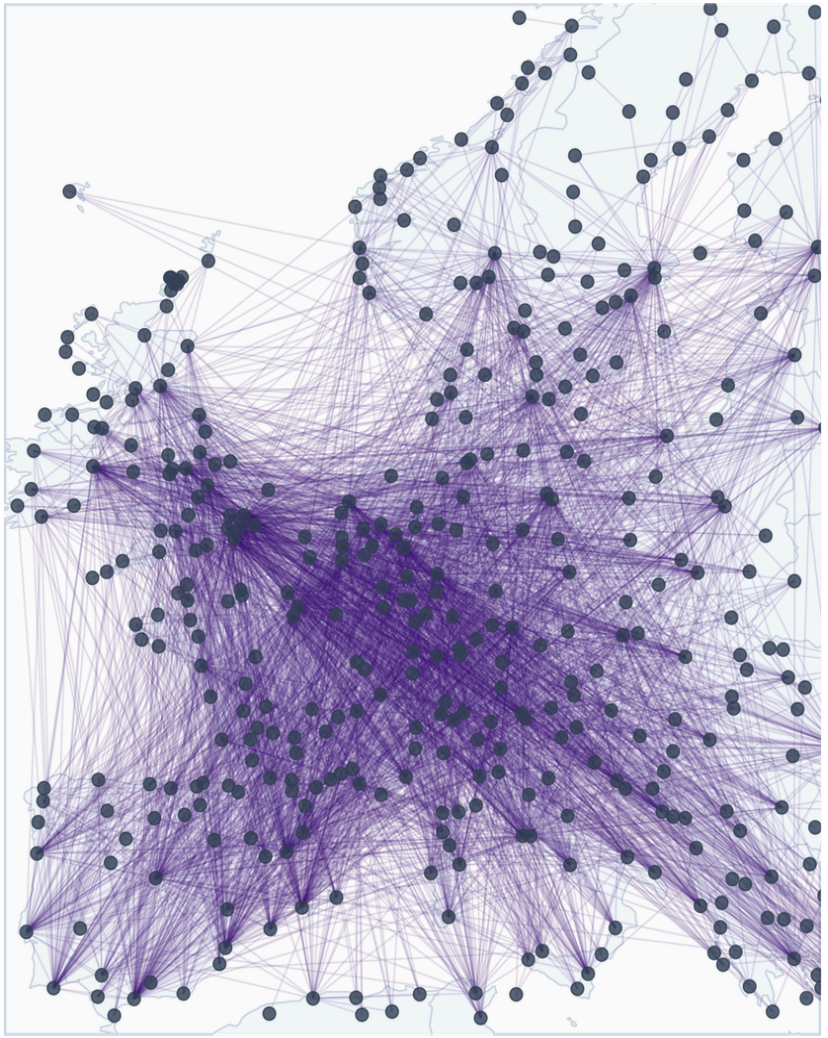

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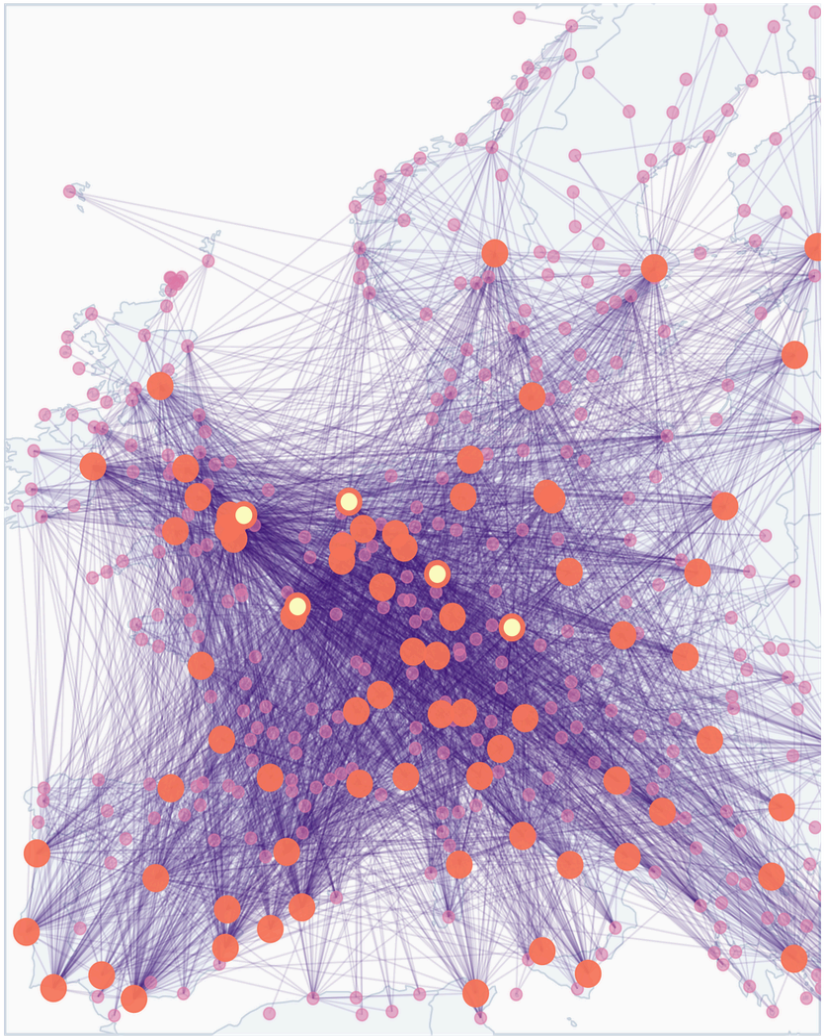

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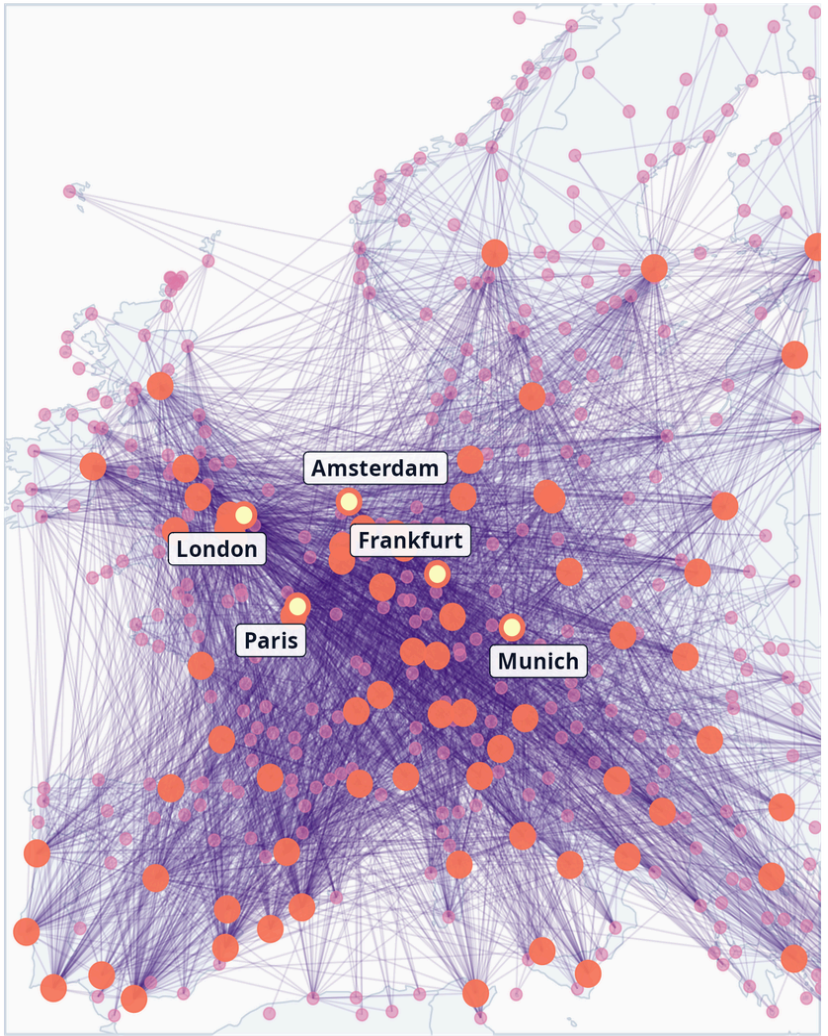

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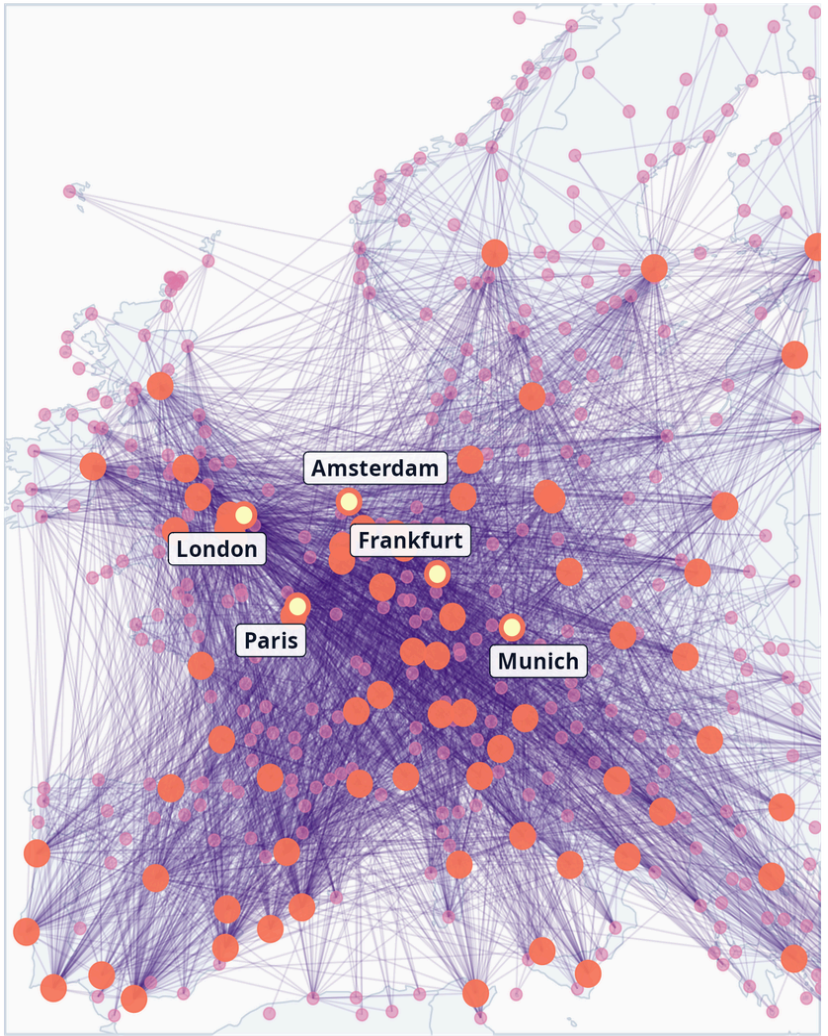
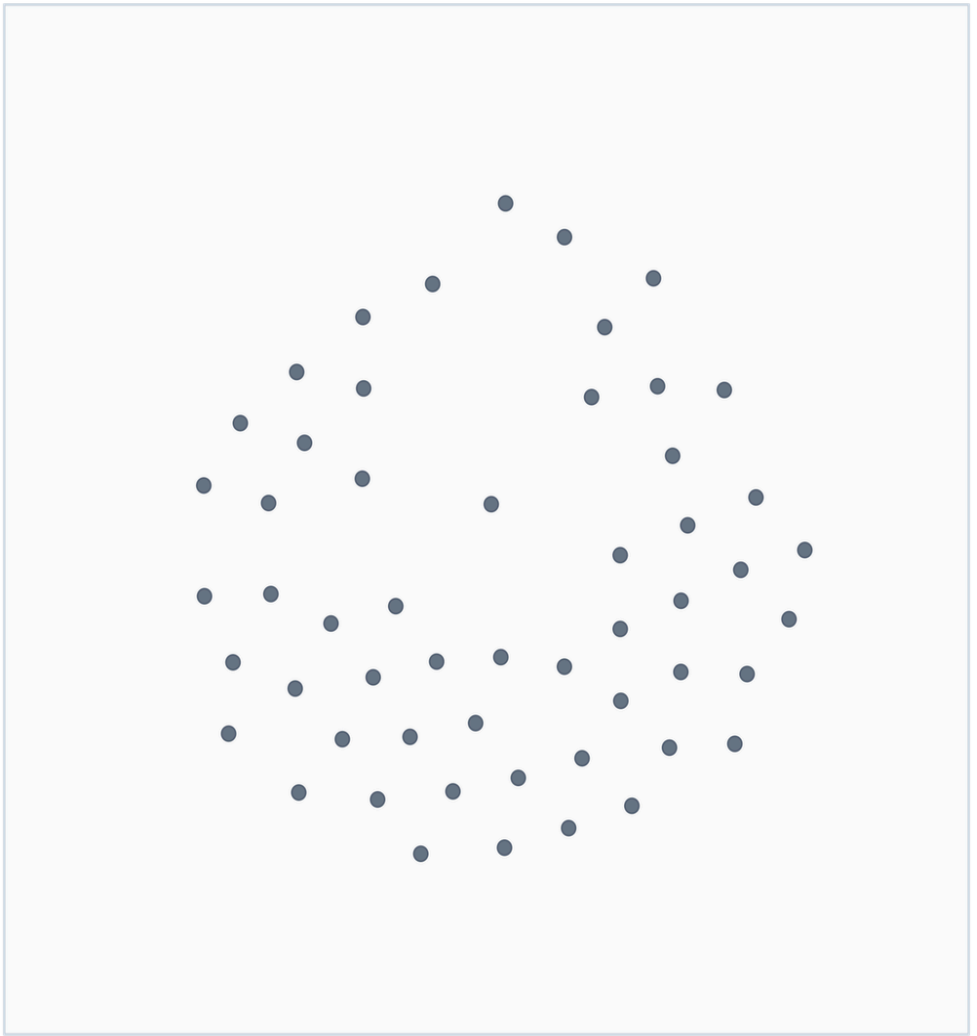
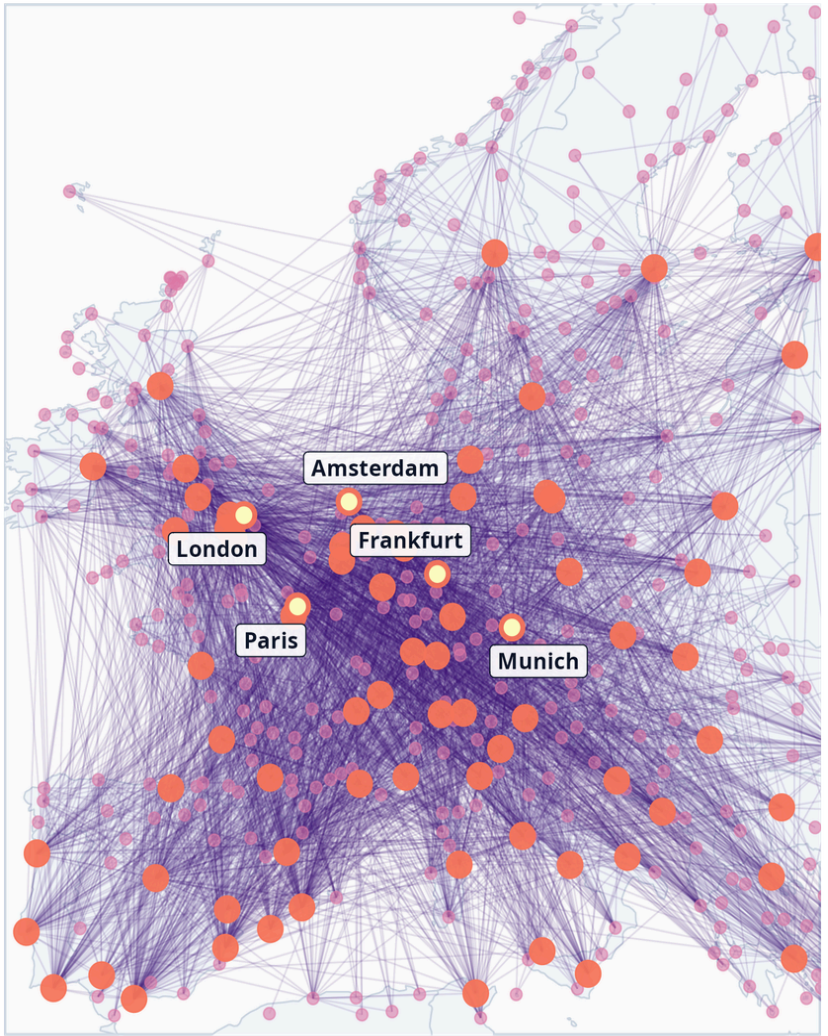


🌸 Meadow Pollinator Sharing



Airports and Meadows

✈️ European Aviation Network



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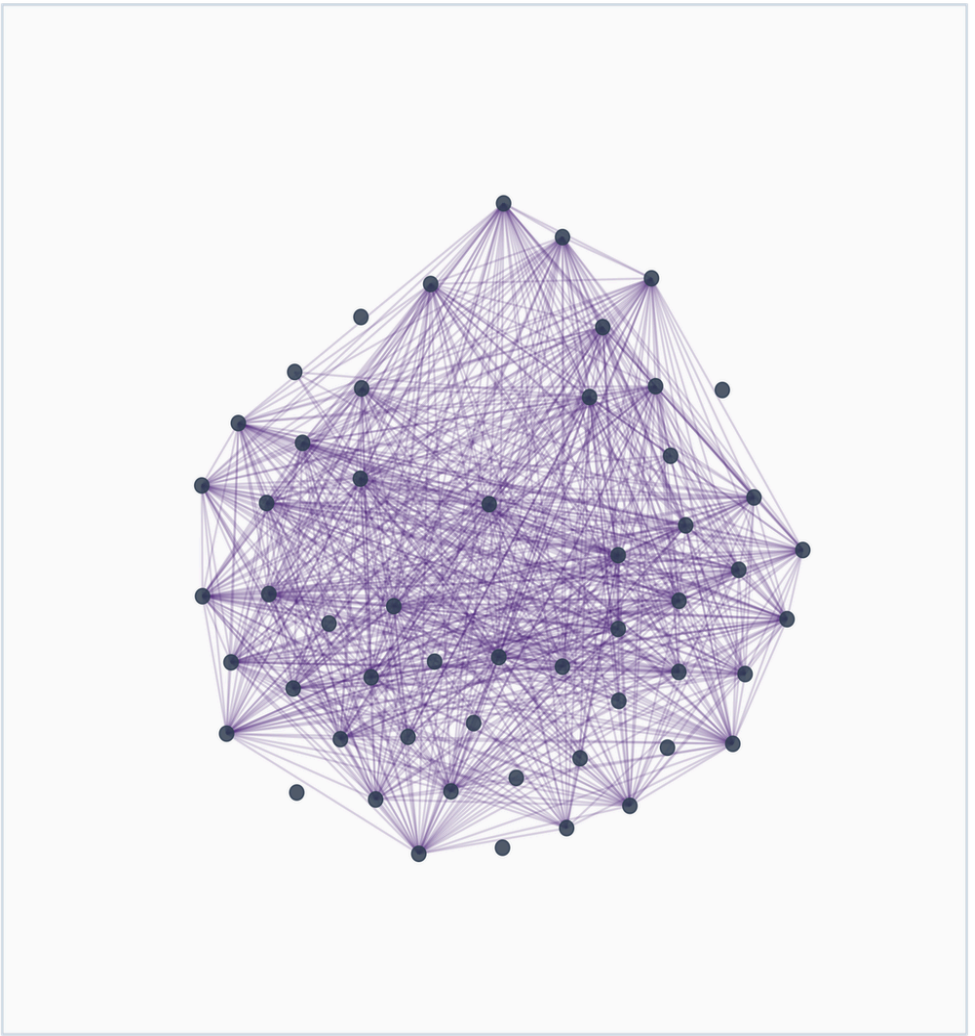

Airports and Meadows

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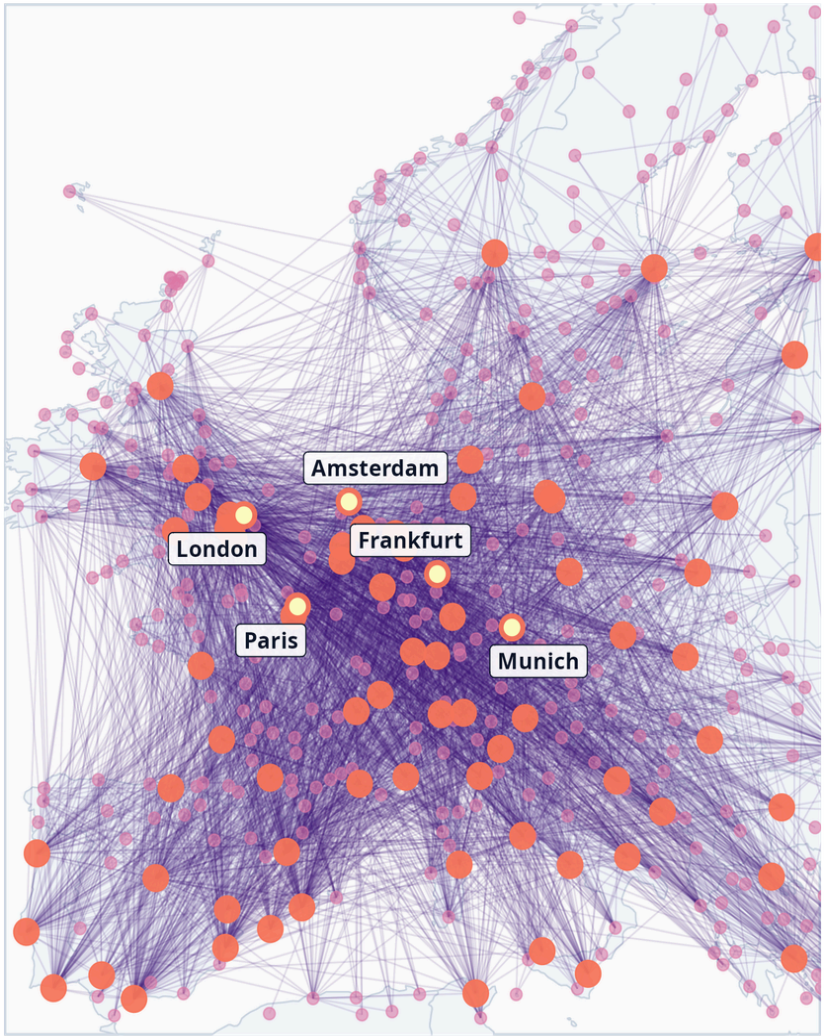

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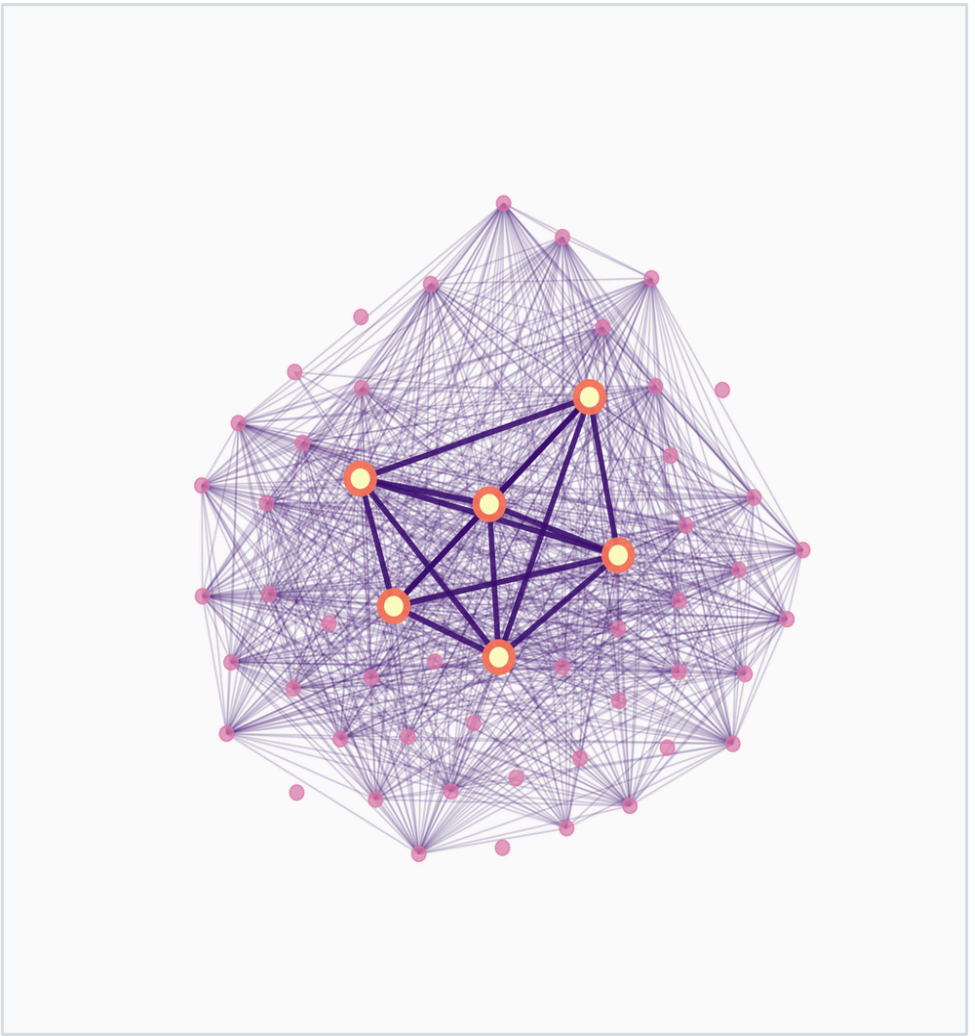
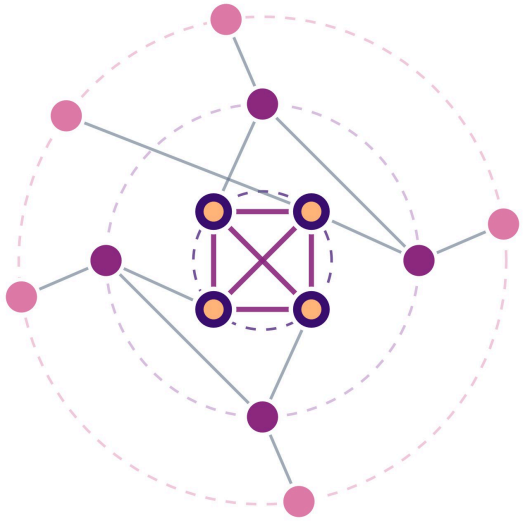


Defining the Core

Defining the Core

Defining the Core

- **Step 1: Intact Web**
  All species (k ≥ 1)



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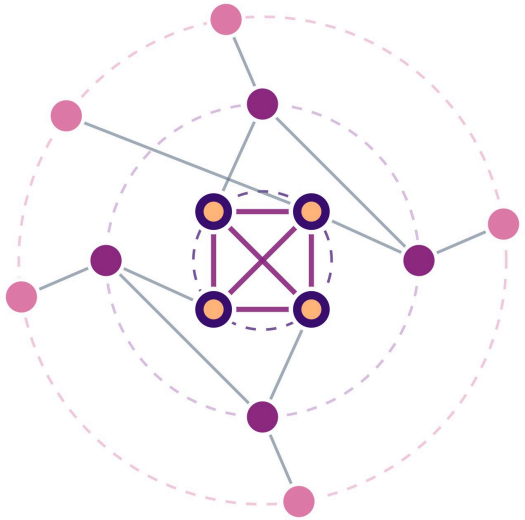
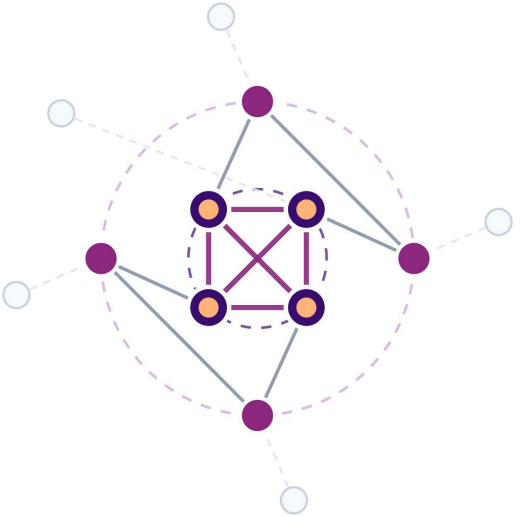
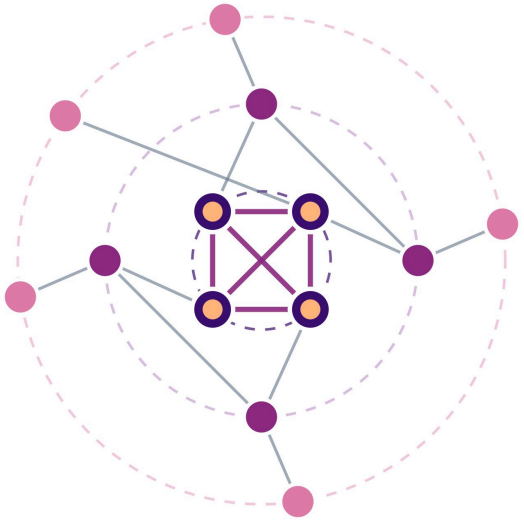


- **Step 2: Iterative Peeling**
  Pruning periphery (k ≥ 2)



Defining the Core

- **Step 1: Intact Web**
  All species ($k \geq 1$)



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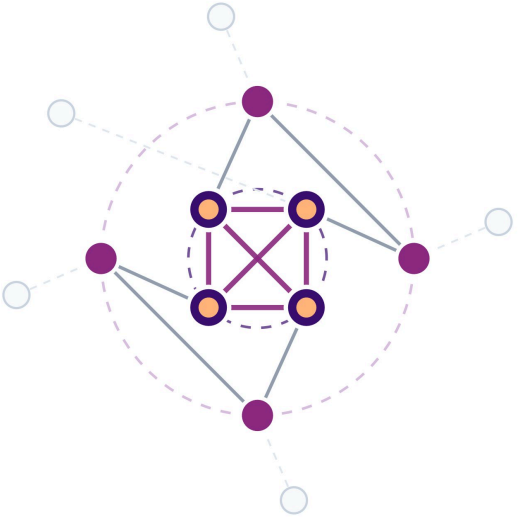
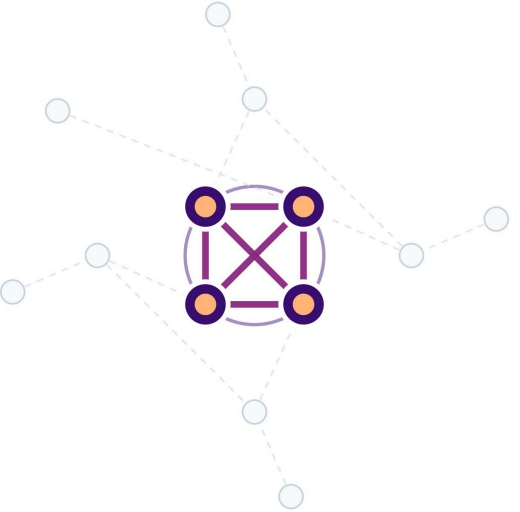
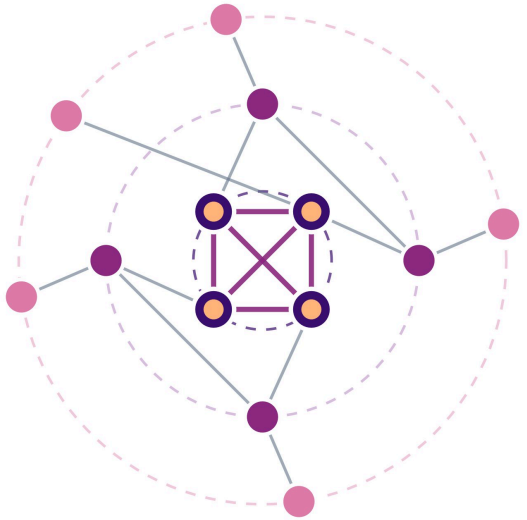


- **Step 3: Invariant Core**
  Mutually linked backbone ($k = k_{max}$)

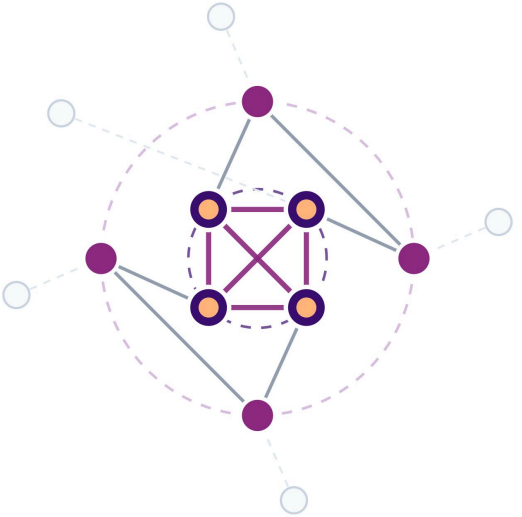


Defining the Core

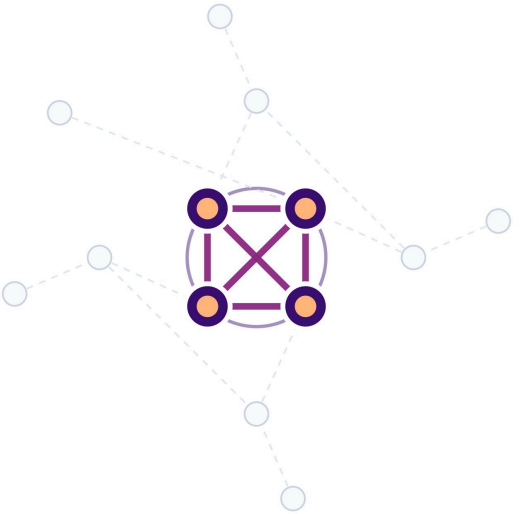
Step 1: Intact Web
All species (k ≥ 1)

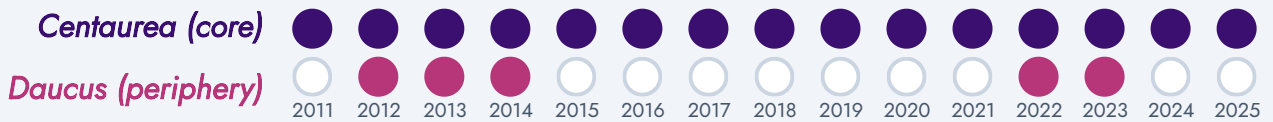


Step 2: Iterative Peeling
Pruning periphery (k ≥ 2)



Step 3: Invariant Core
Mutually linked backbone (k = k_{max})



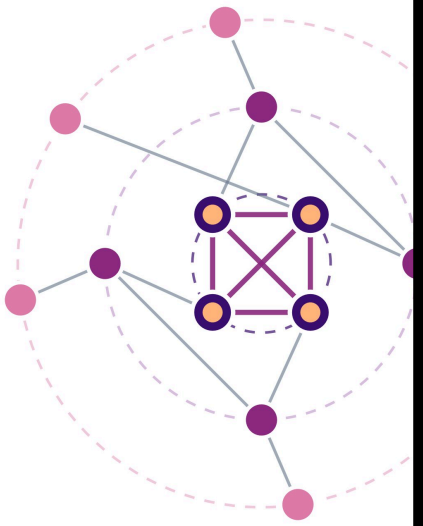


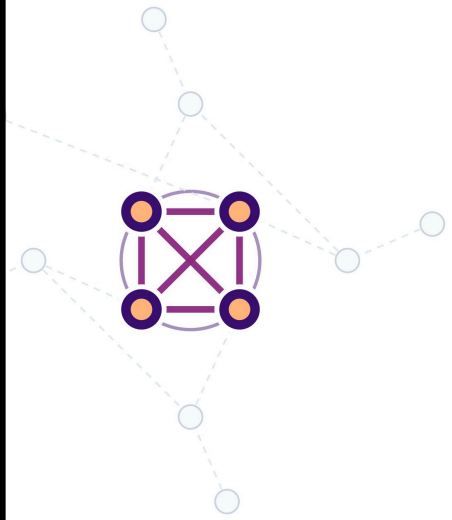
Defining the Core

- **Step 1: Intact Web**
  All species (k ≥ 1)

- **Step 2: Iterative Peeling**

- **Step 3: Invariant Core**
  tually linked backbone (k = k_{max})





Centaurea

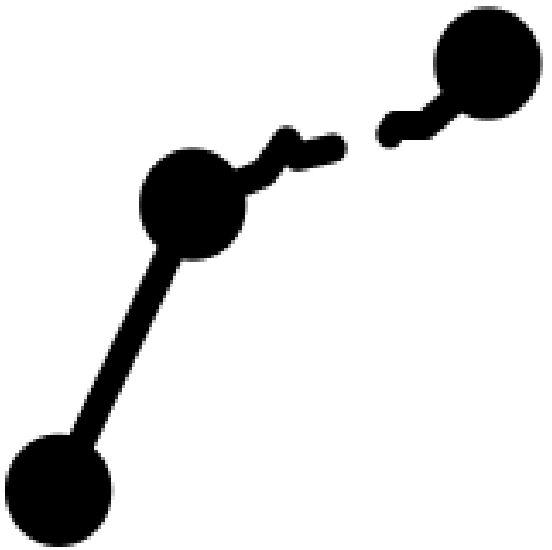
Daucus (peri

25

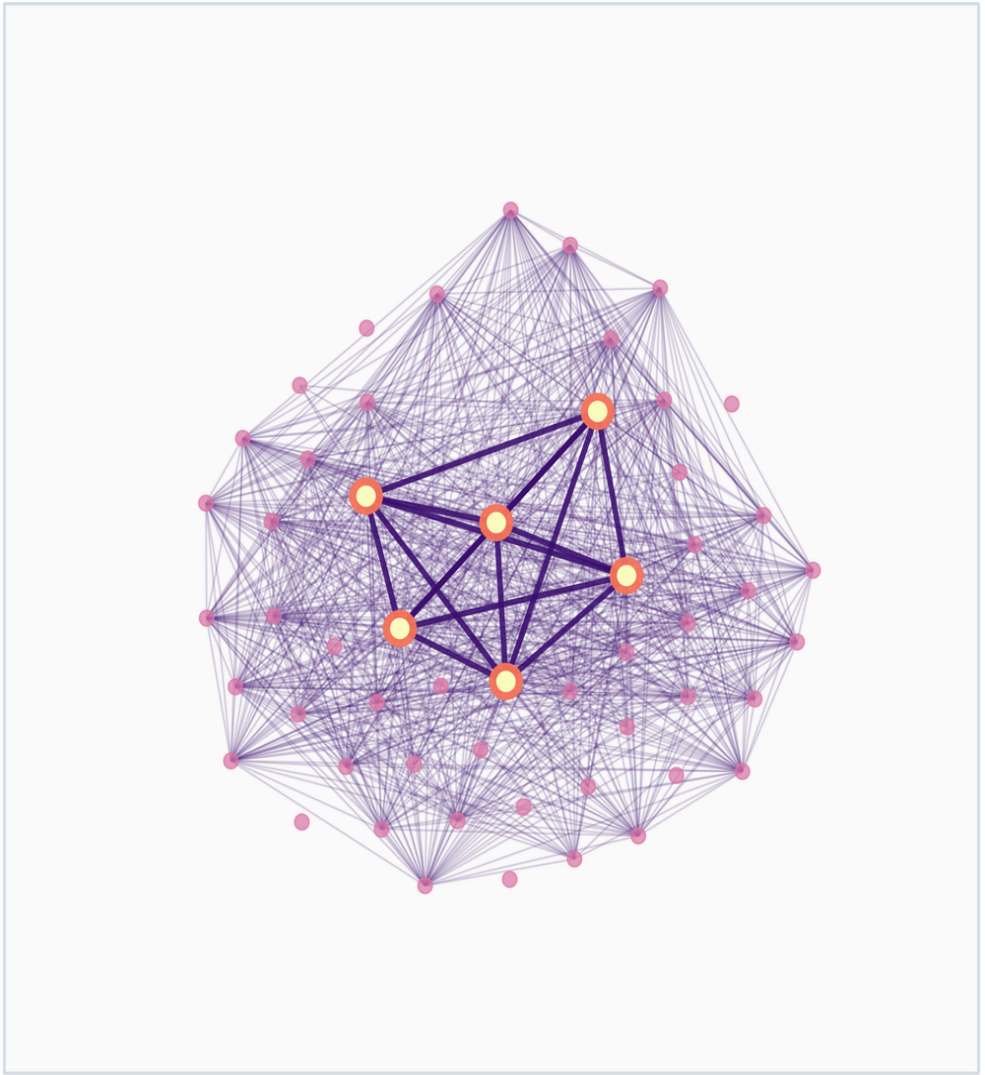
- central that year
  15 / 15 vs 5 / 15 years

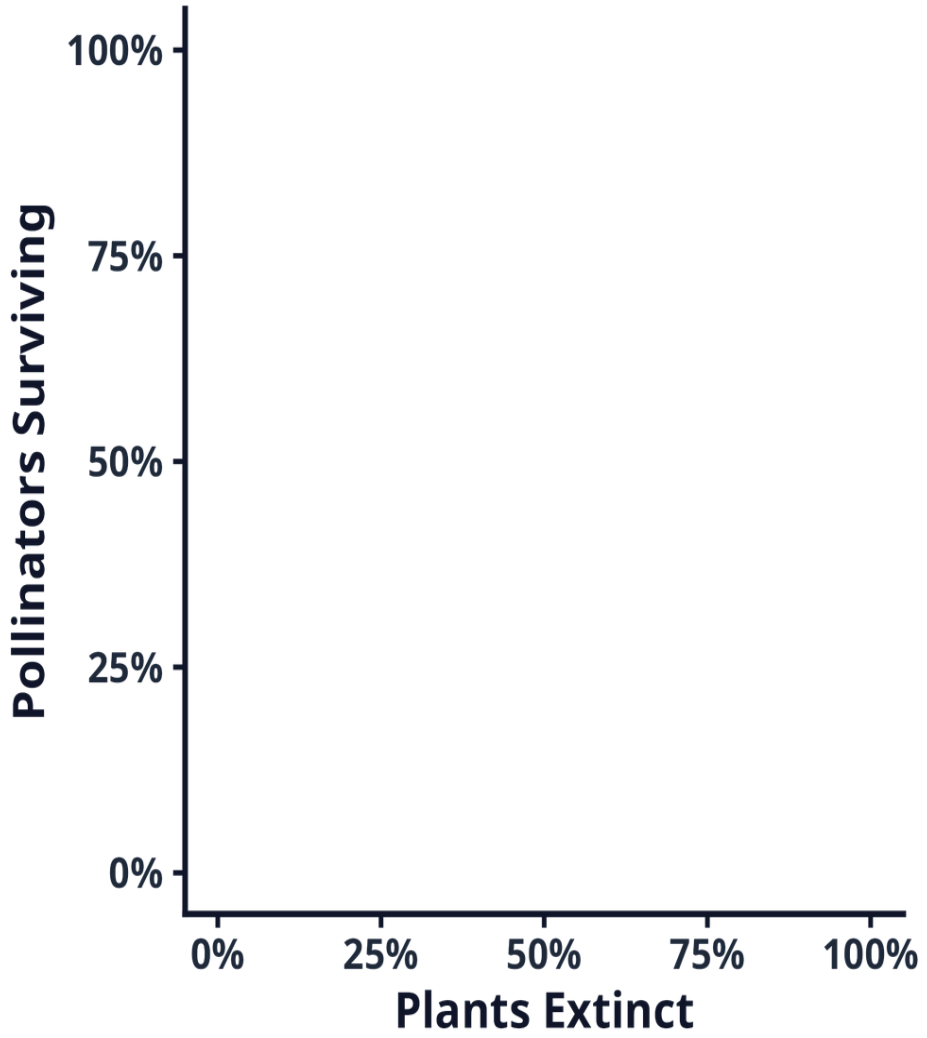
Robustness

Robustness

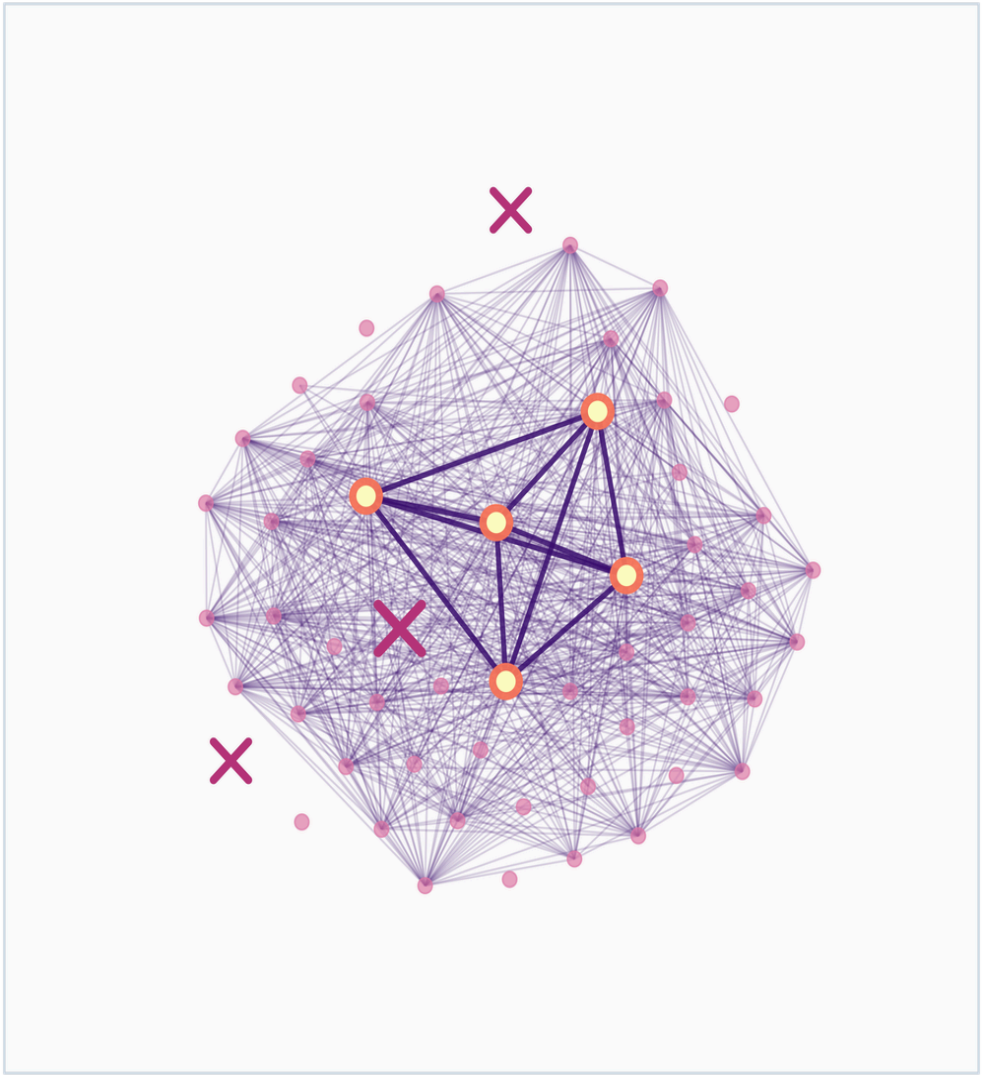


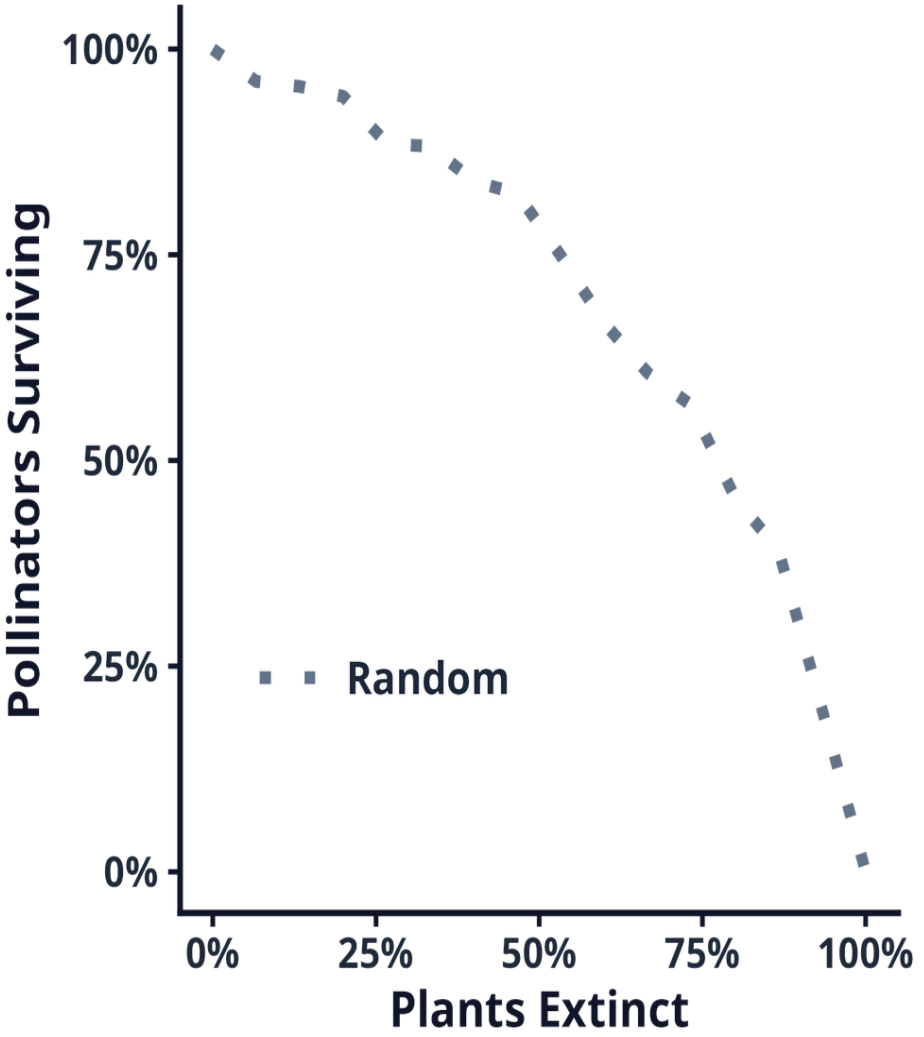
Robustness



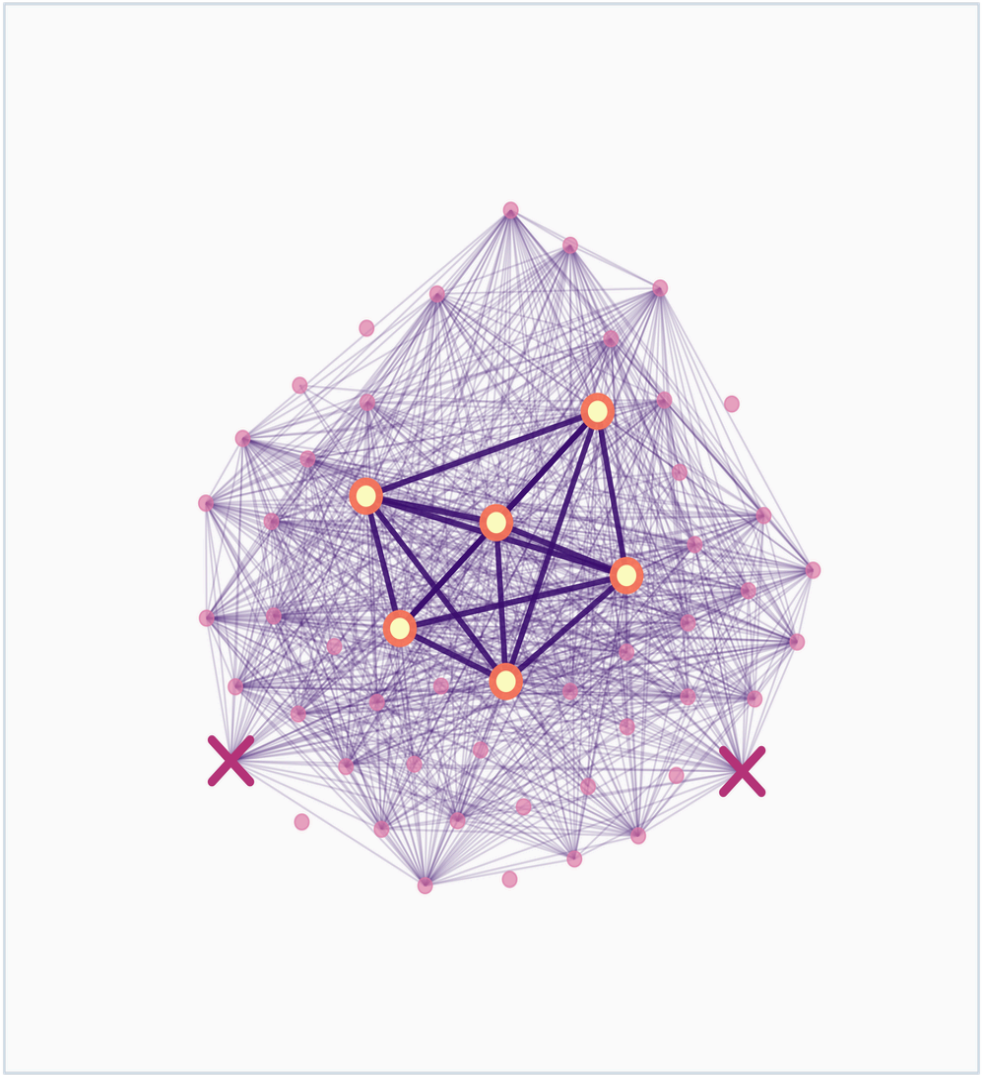

100%
75%
50%
25%
0%
Pollinators Surviving
0%
25%
50%
75%
100%
Plants Extinct

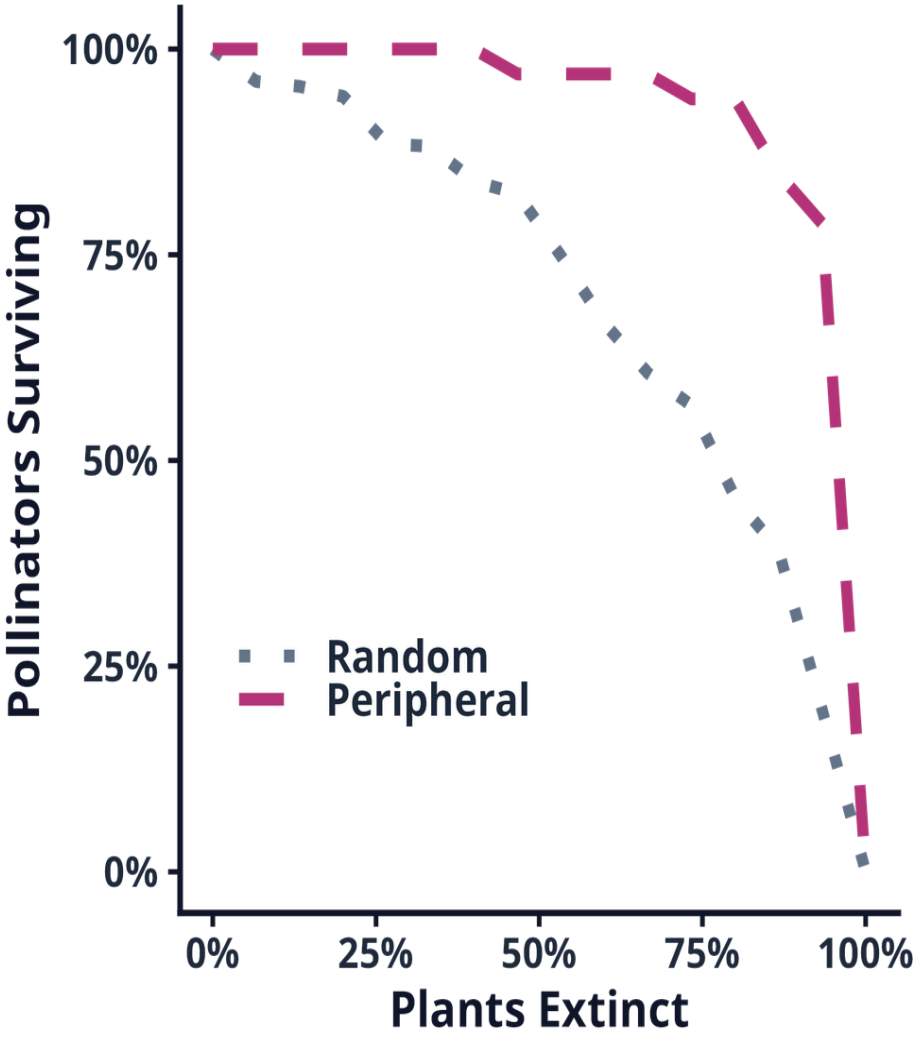
Robustness



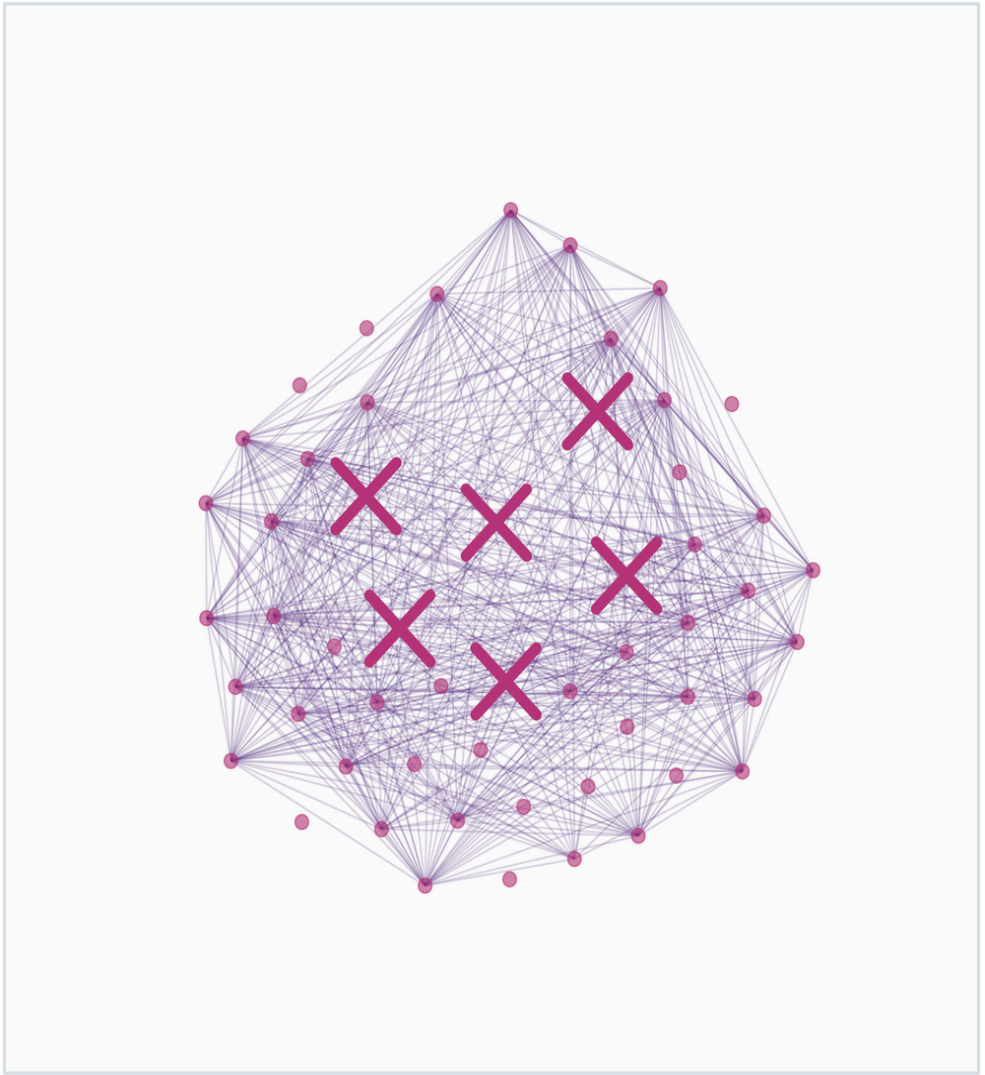

Pollinators Surviving
100%
75%
50%
25%
0%
Random
0%
25%
50%
75%
100%
Plants Extinct

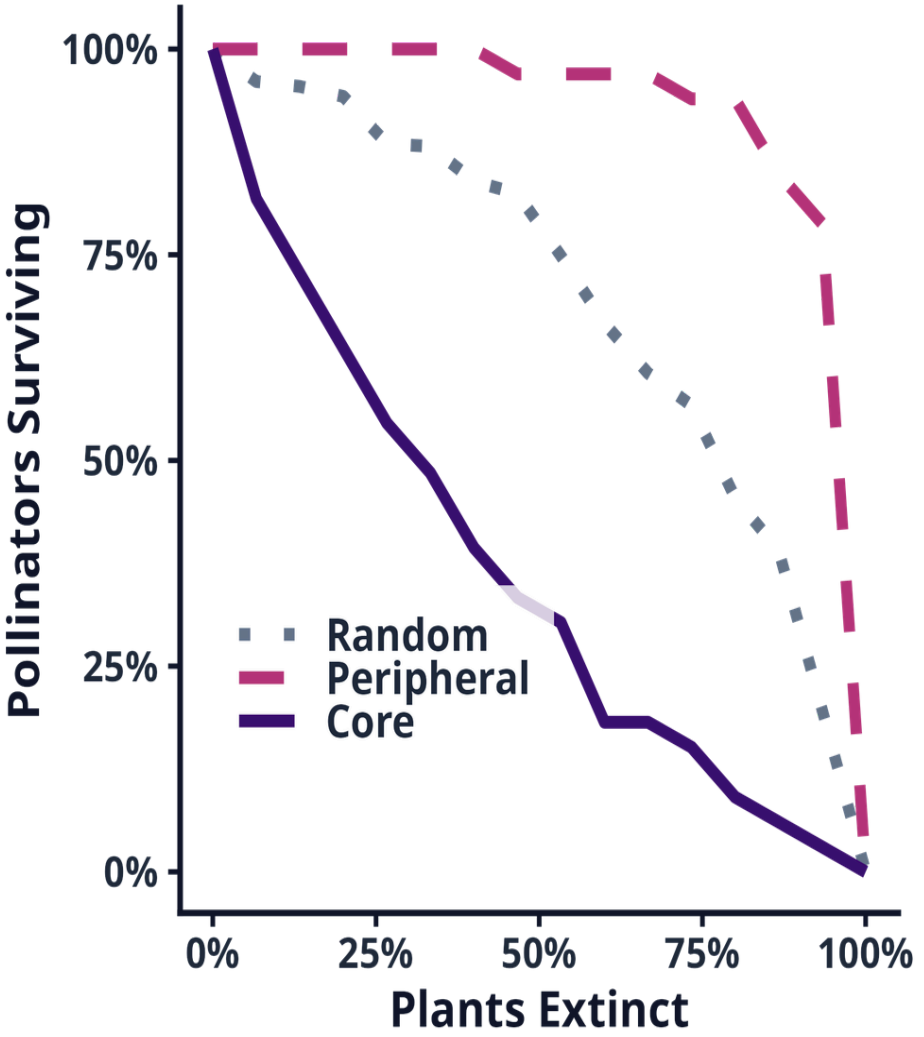
Robustness




Pollinators Surviving
100%
75%
50%
25%
0%
Random
Peripheral
0%
25%
50%
75%
100%
Plants Extinct

Robustness

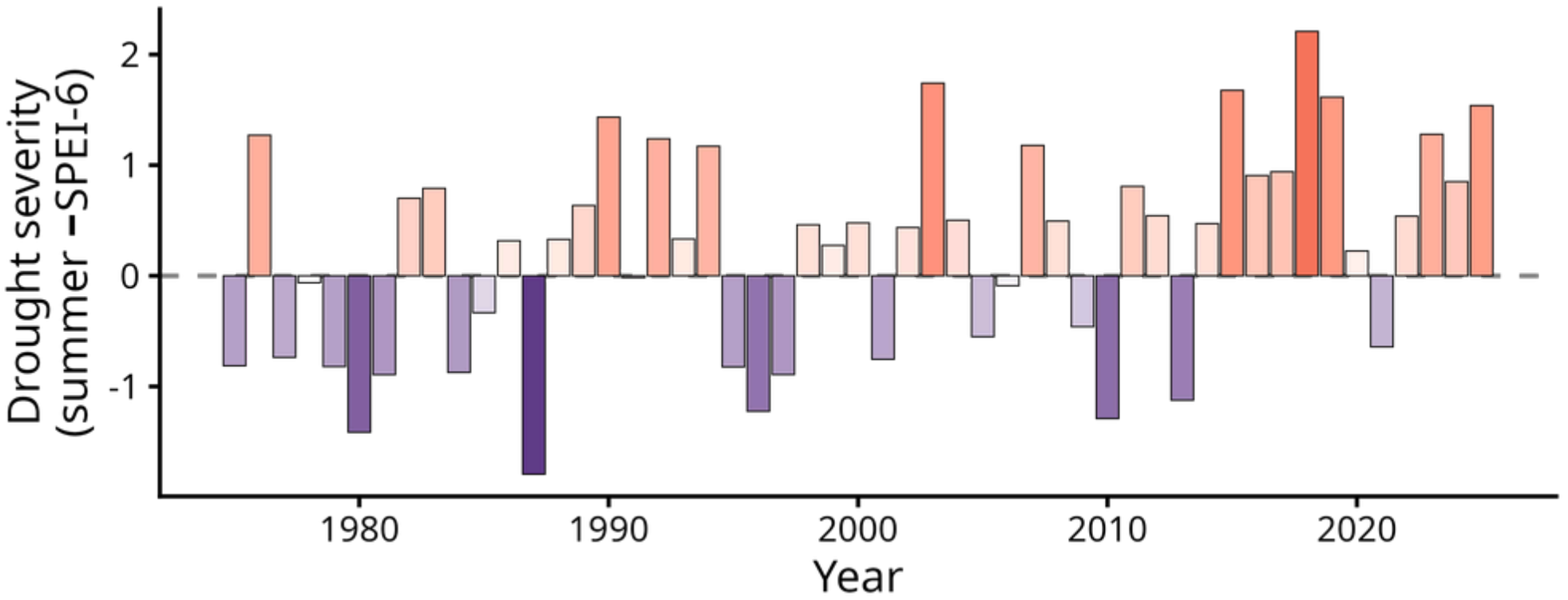



Pollinators Surviving
100%
75%
50%
25%
0%
0%
25%
50%
75%
100%
Plants Extinct
Random
Peripheral
Core

Drought

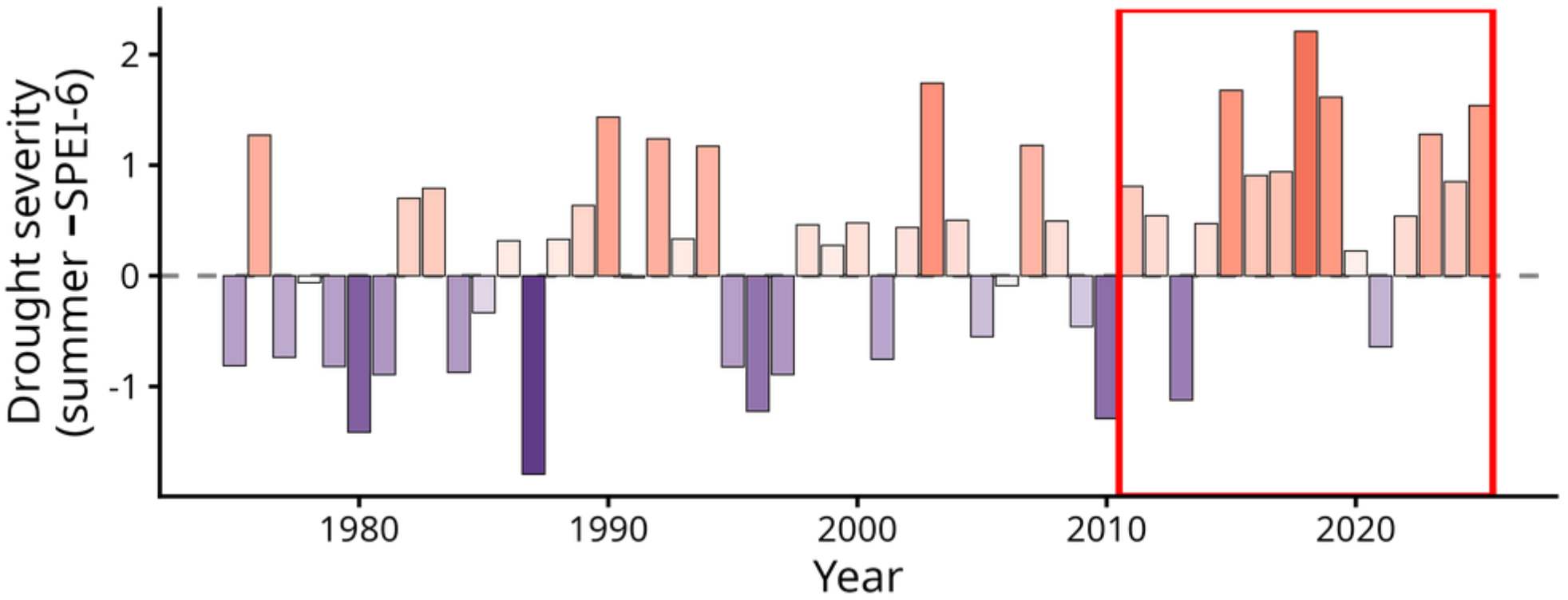
Drought

Drought



Data: TerraClimate (Abatzoglou et al., 2018), Handrkov locality (1975–2025)

Drought



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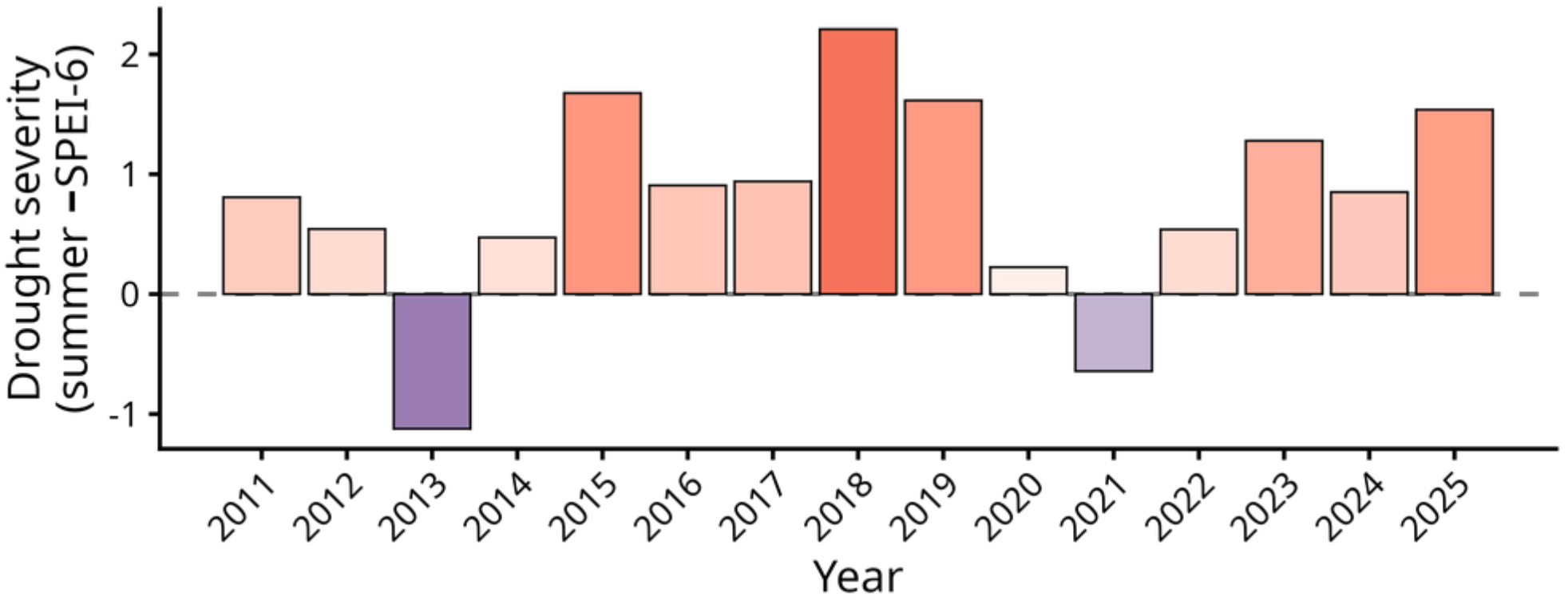
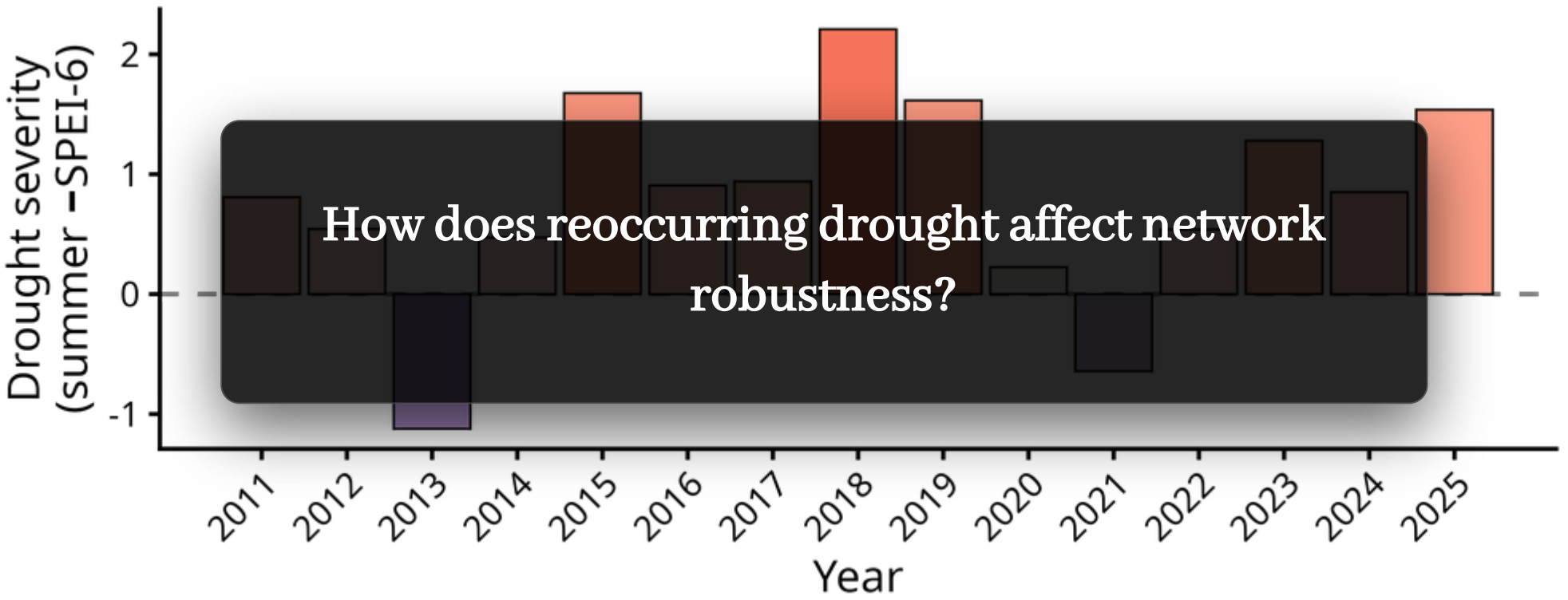
Drought



Data: TerraClimate (Abatzoglou et al., 2018), Handrkov locality (2011–2025)

Drought



Data: TerraClimate (Abatzoglou et al., 2018), Handrkov locality (2011–2025)

Methods - the locality

Methods - the locality

Czech Republic

Methods - the locality

Czech Republic

Rich in flowers

Methods - the locality

Czech Republic

Rich in flowers

15 years of data collection (2011–2025)

Methods - permanent plots

Methods - permanent plots

plots are marked

Methods - permanent plots

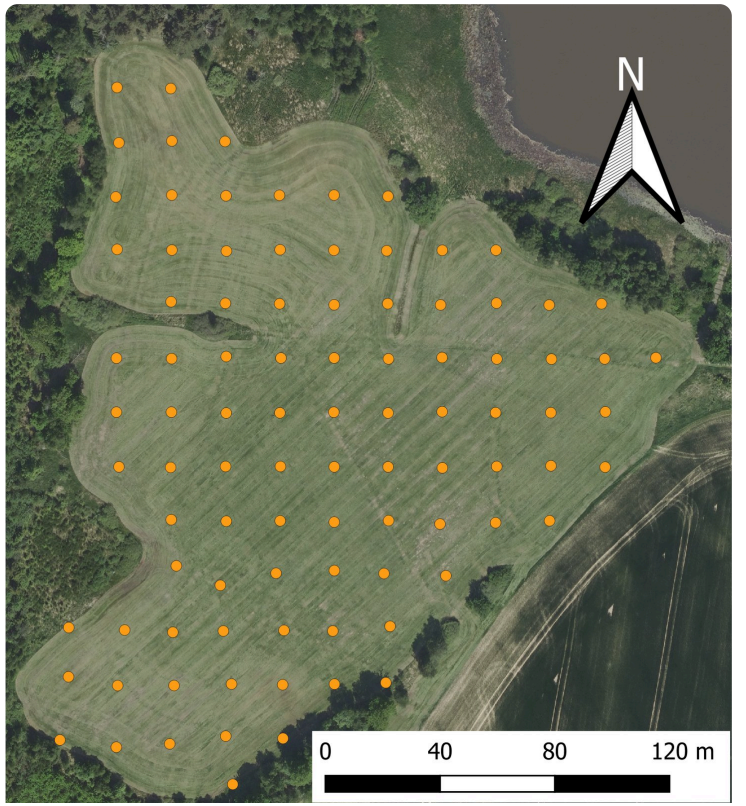
plots are marked

recovered every year

Methods - permanent plots

plots are marked

recovered every year

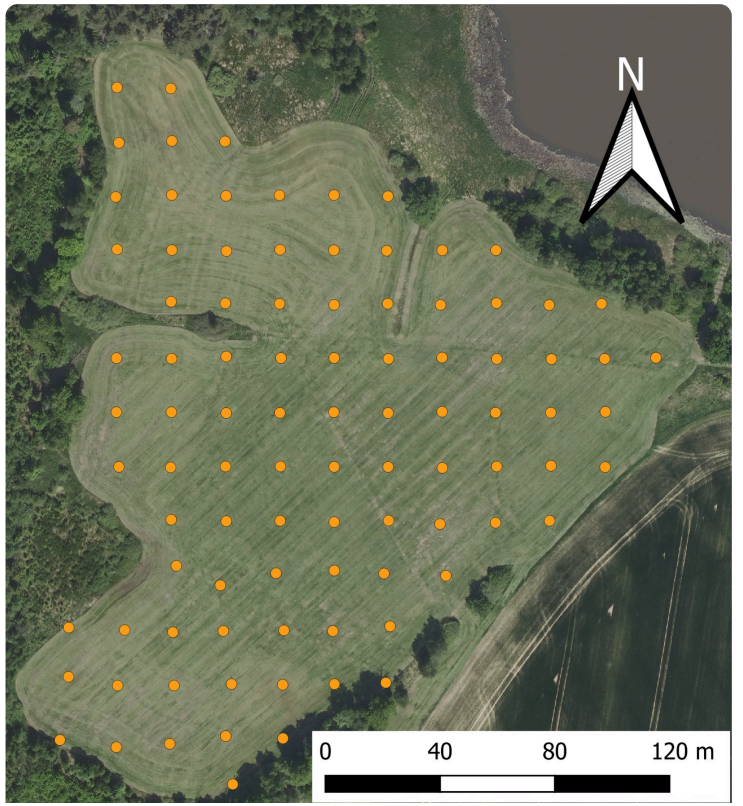


93 plots in total

Methods - permanent plots

plots are marked

recovered every year



93 plots in total

- sampled once per year in August

Methods - plant-pollinator survey

Methods - plant-pollinator survey

Plant survey

Methods - plant-pollinator survey

Plant survey

Pollinator survey

Methods - plant survey

Methods - plant survey

Methods - plant survey

- 4 x 4 meter squares
- Stalks of flowering plants are counted or scored presence/absence in subplots
- 2 times per sampling period

Methods - pollinator survey

Methods - pollinator survey

Methods - pollinator survey

- “Snapshot” of plant-pollinator interaction
- ~28 surveys per plot per year
- 42 378 surveys in total
- 68 460 interactions recorded
- 176 pollinator taxa × 55 plant species

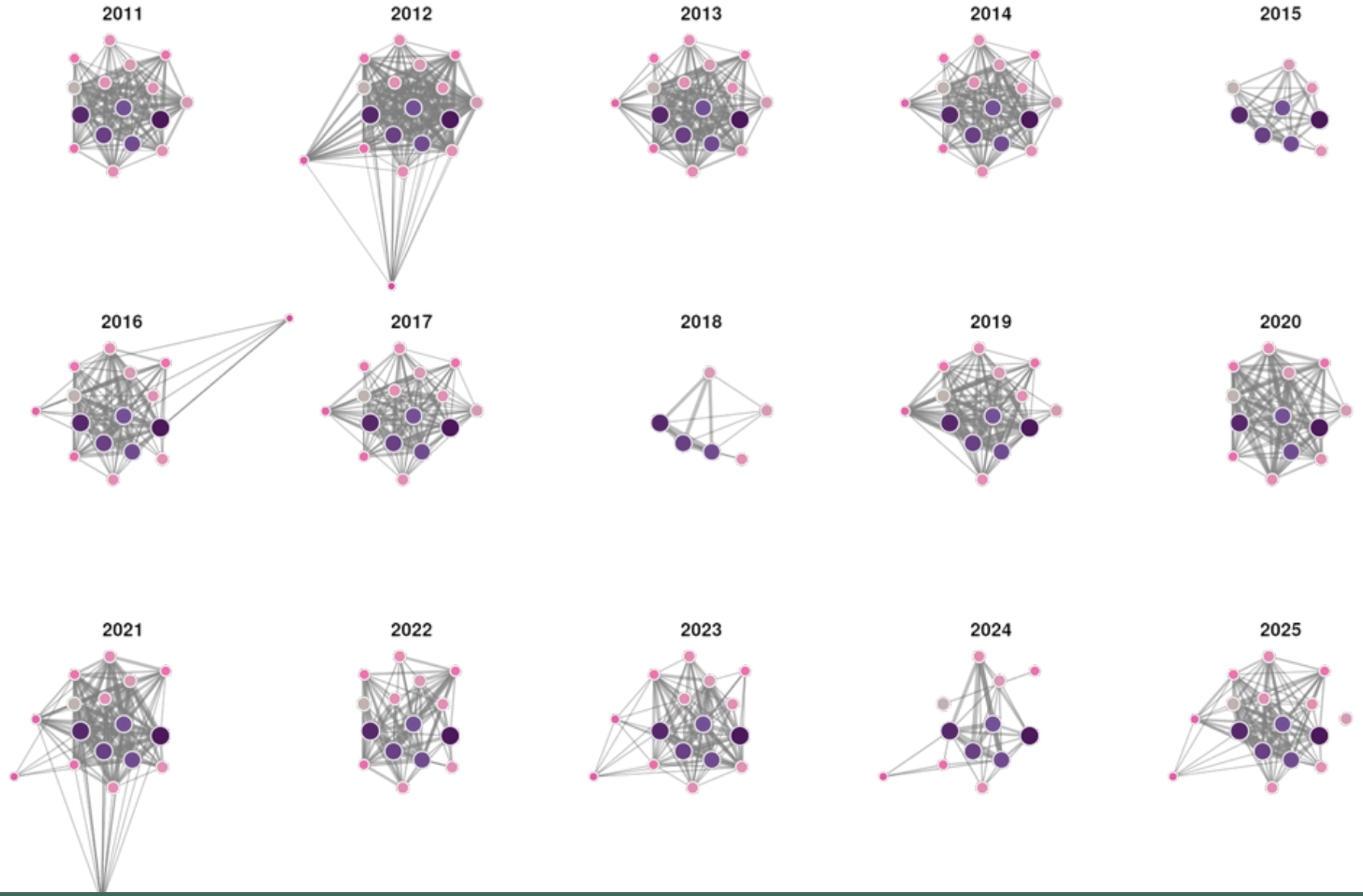
Methods - plant-pollinator survey

Methods - plant-pollinator survey

Methods - plant-pollinator survey

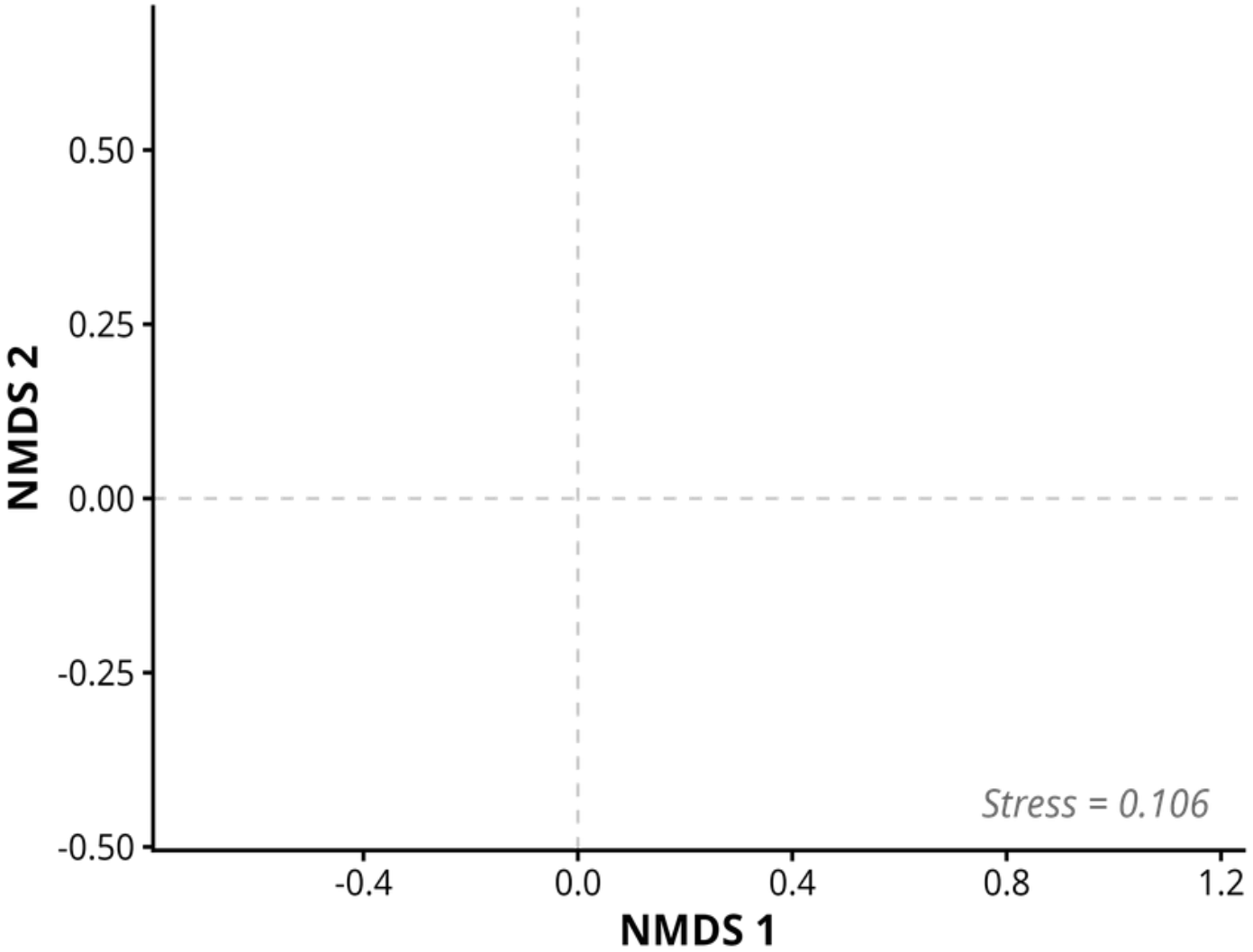
Methods - plant-pollinator survey

Networks


2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025

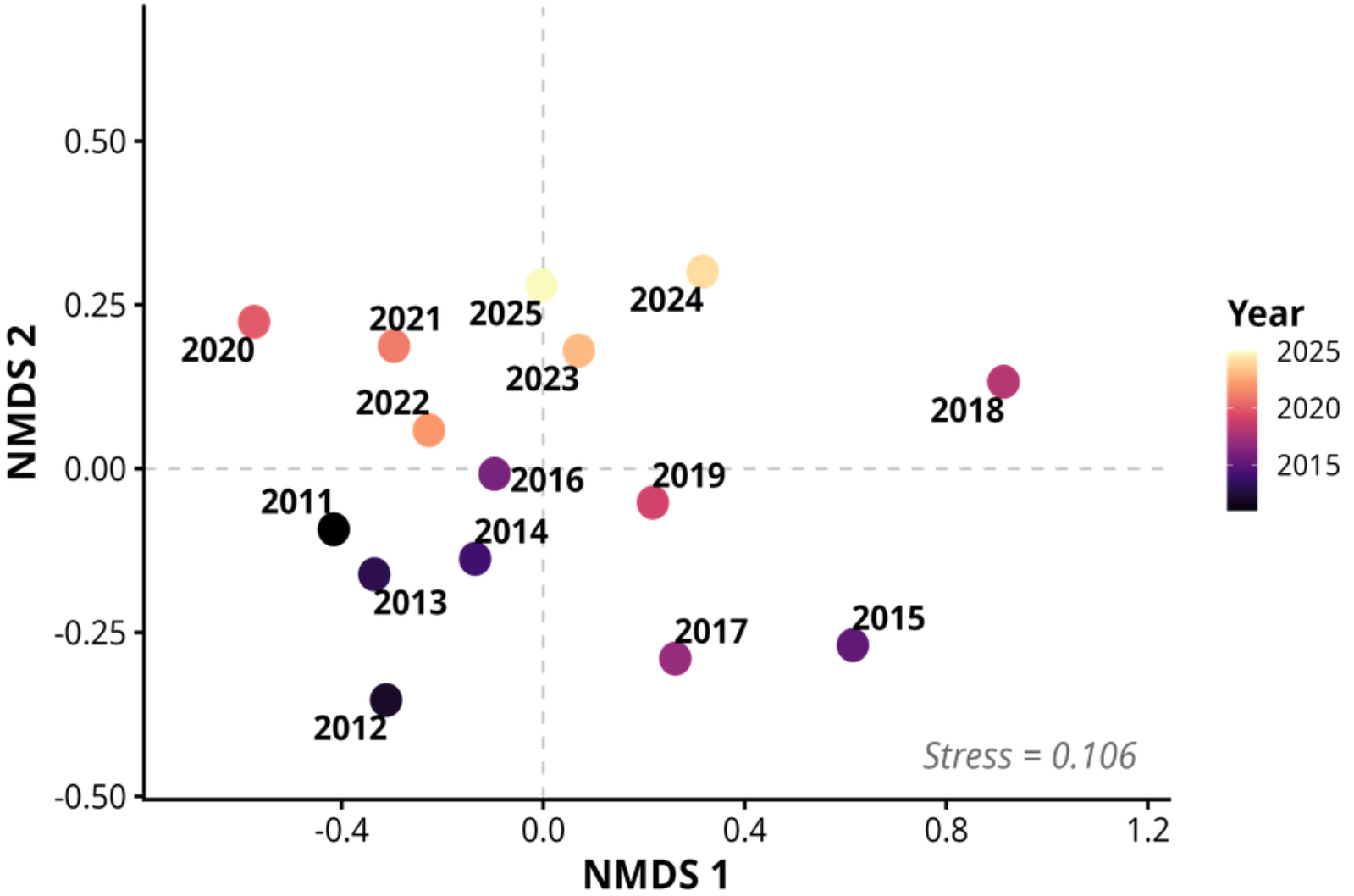
Year-to-Year Change

Pollinator sharing: Müller's index (PAC)
NMDS, Bray–Curtis



Year-to-Year Change

Pollinator sharing: Müller's index (PAC)
NMDS, Bray–Curtis



Year-to-Year Change

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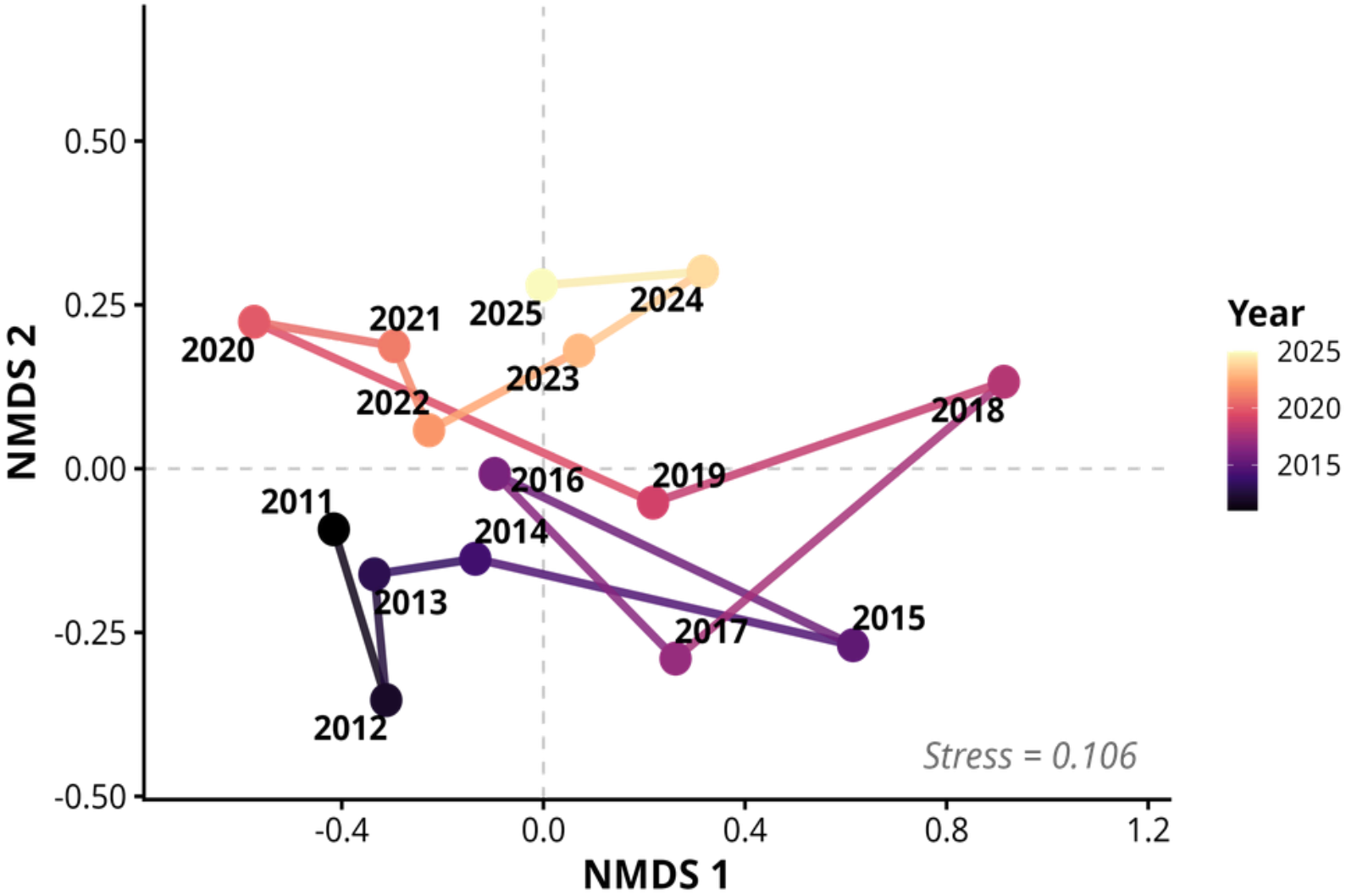

Year-to-Year Change

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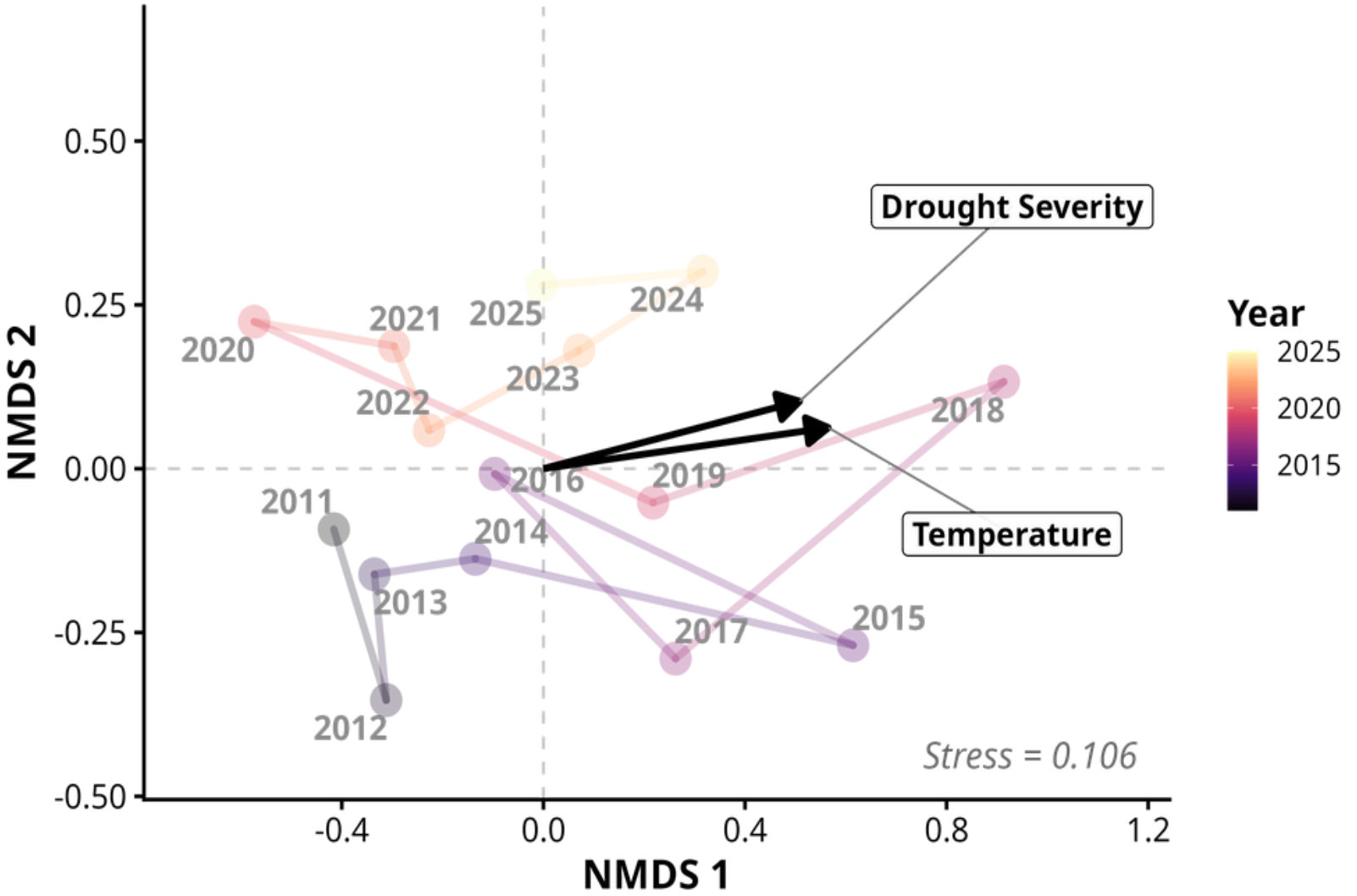

Year-to-Year Change

Pollinator sharing: Müller's index (PAC)
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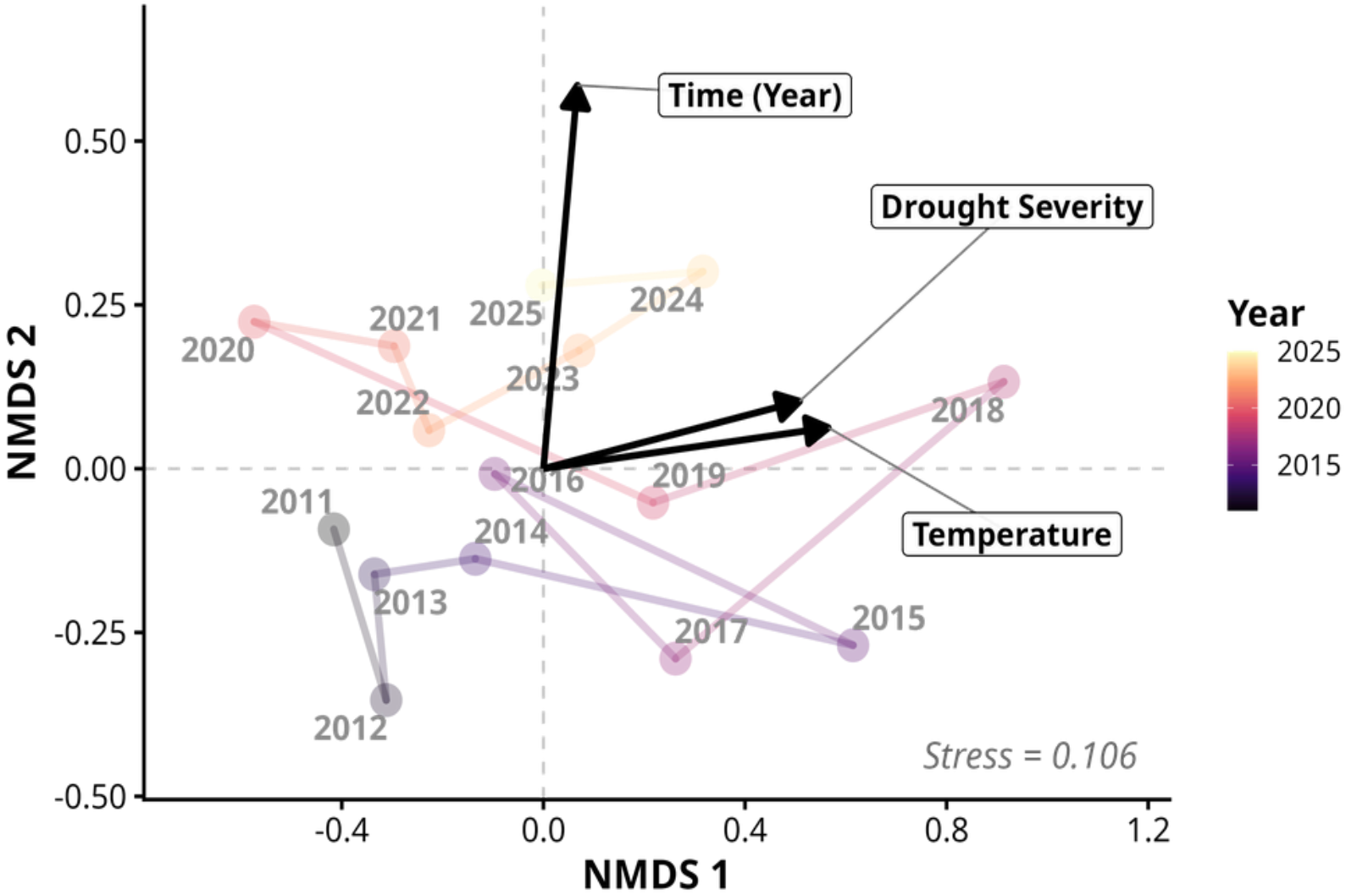
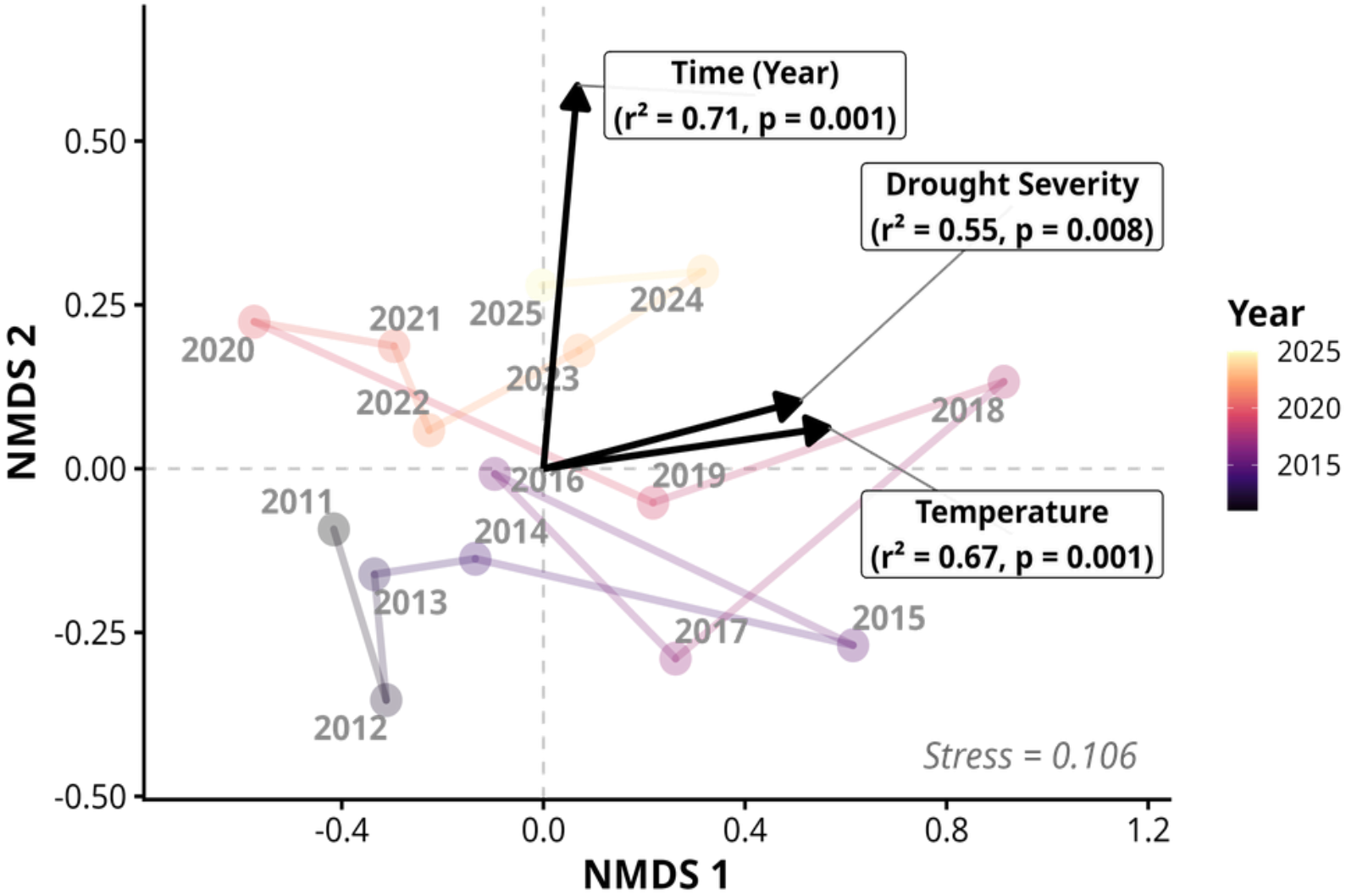


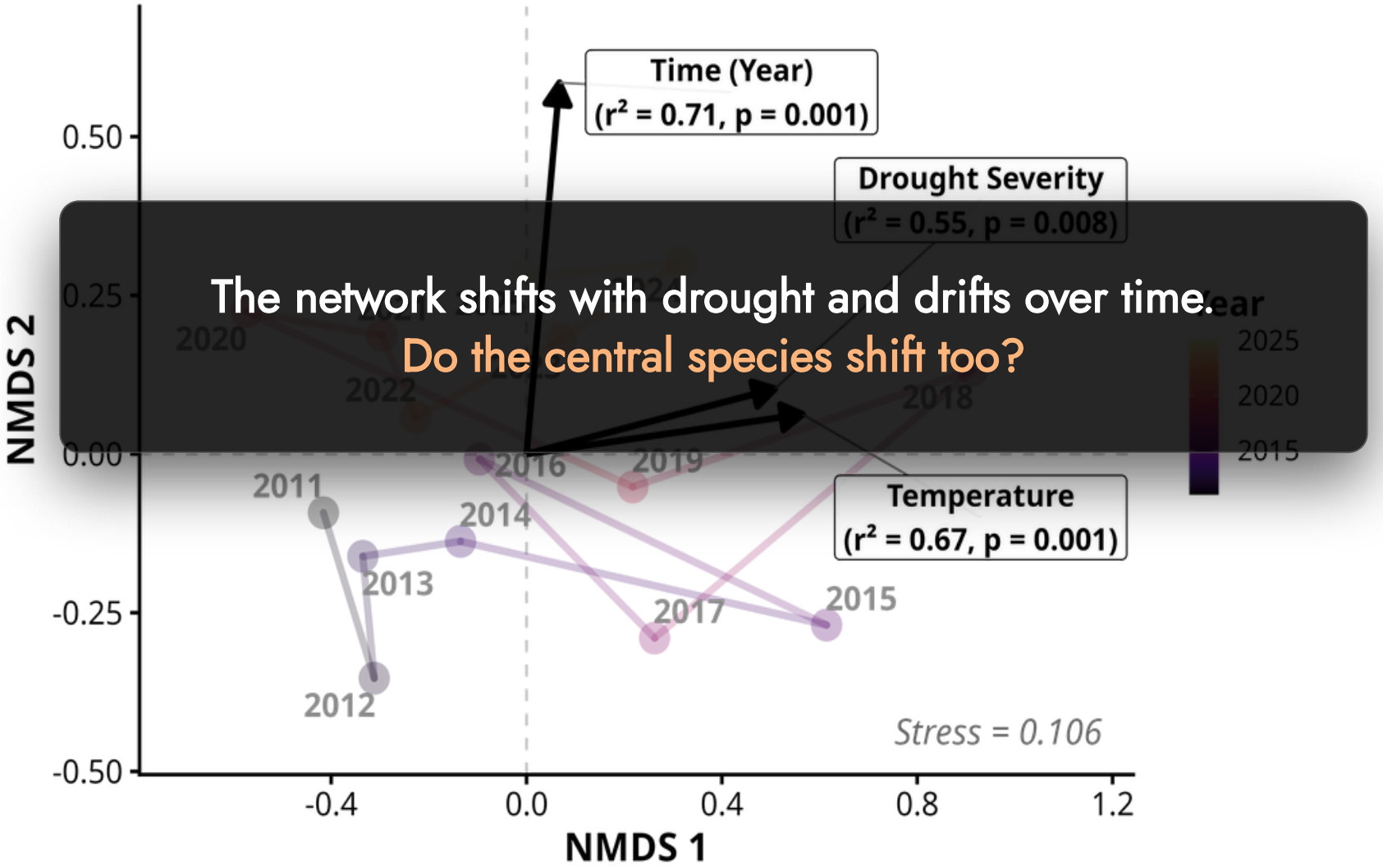
Year-to-Year Change

Pollinator sharing: Müller's index (PAC)
NMDS, Bray–Curtis



Year-to-Year Change

Pollinator sharing: Müller's index (PAC)
NMDS, Bray–Curtis



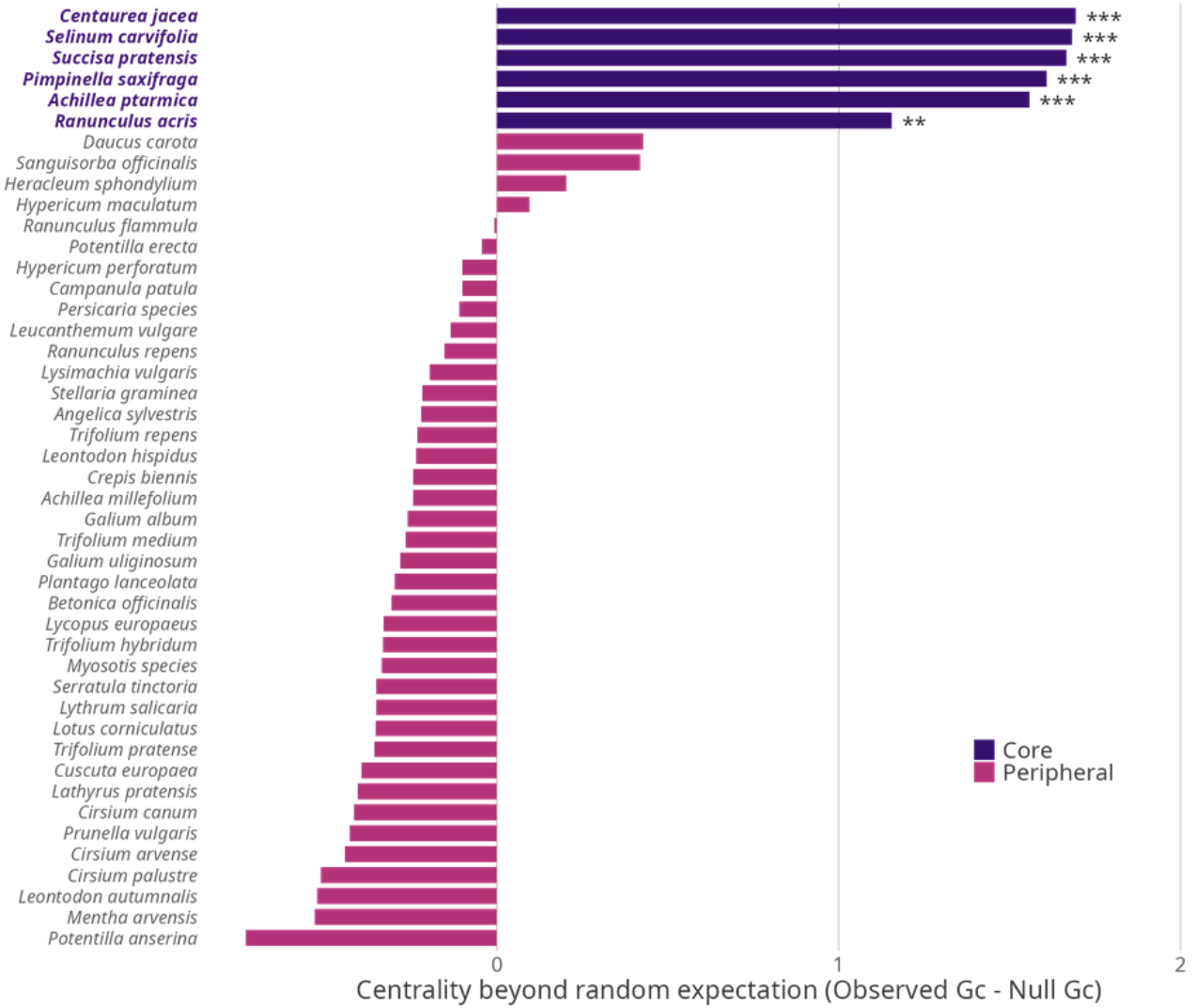
The Core

Centaurea jacea

Selinum carvifolia

Succisa pratensis

Centrality (Gc): number of pollinator taxa visiting the plant, vs. the year's average (Dáttilo et al. 2013)



Pimpinella saxifraga

Achillea ptarmica

Ranunculus acris

The Core

Centrality (Gc): number of pollinator taxa visiting the plant, vs. the year's average (Dáttilo et al. 2013)

Centaurea jacea

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Succisa pratensis

Pimpinella saxifraga

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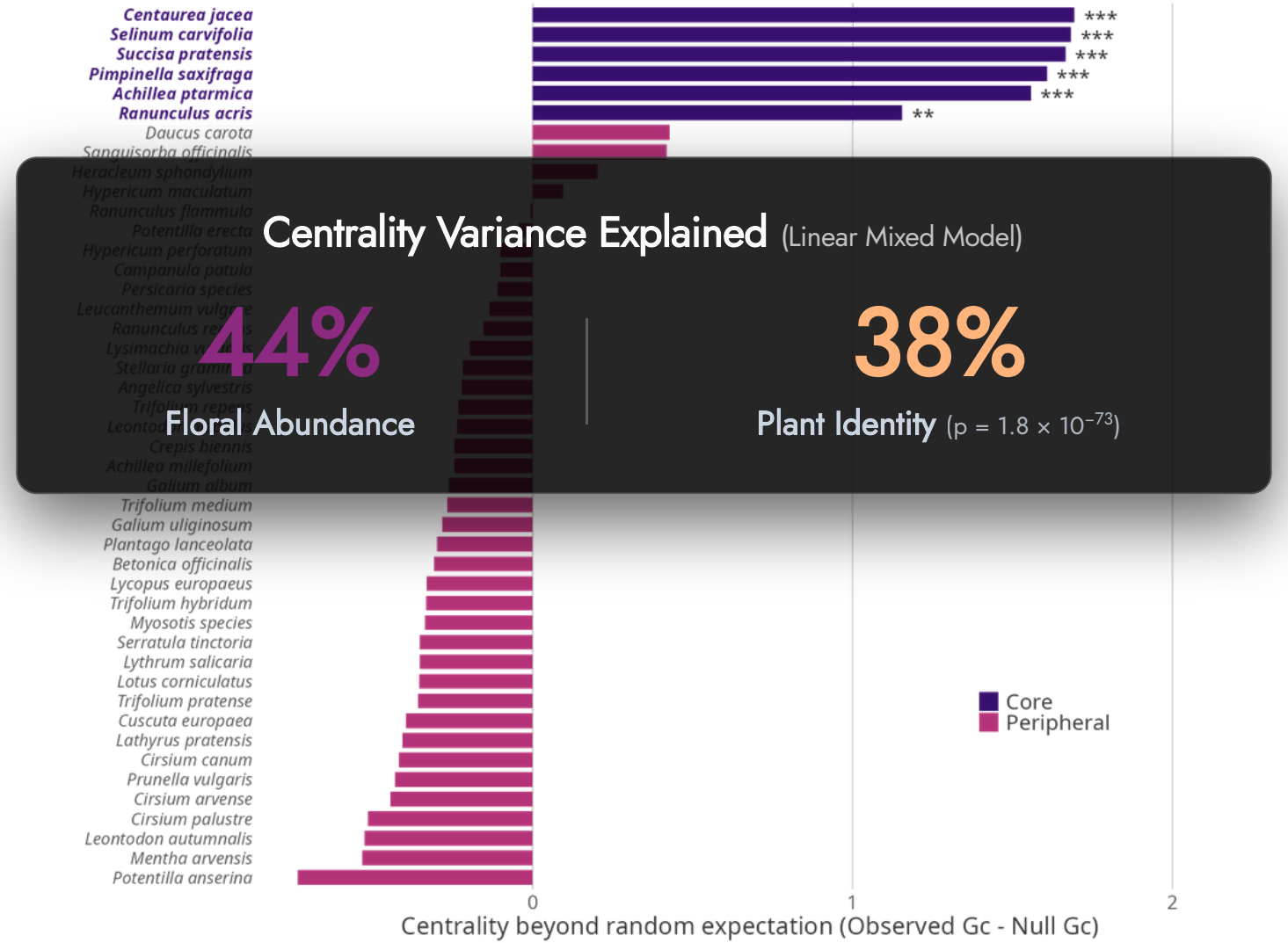
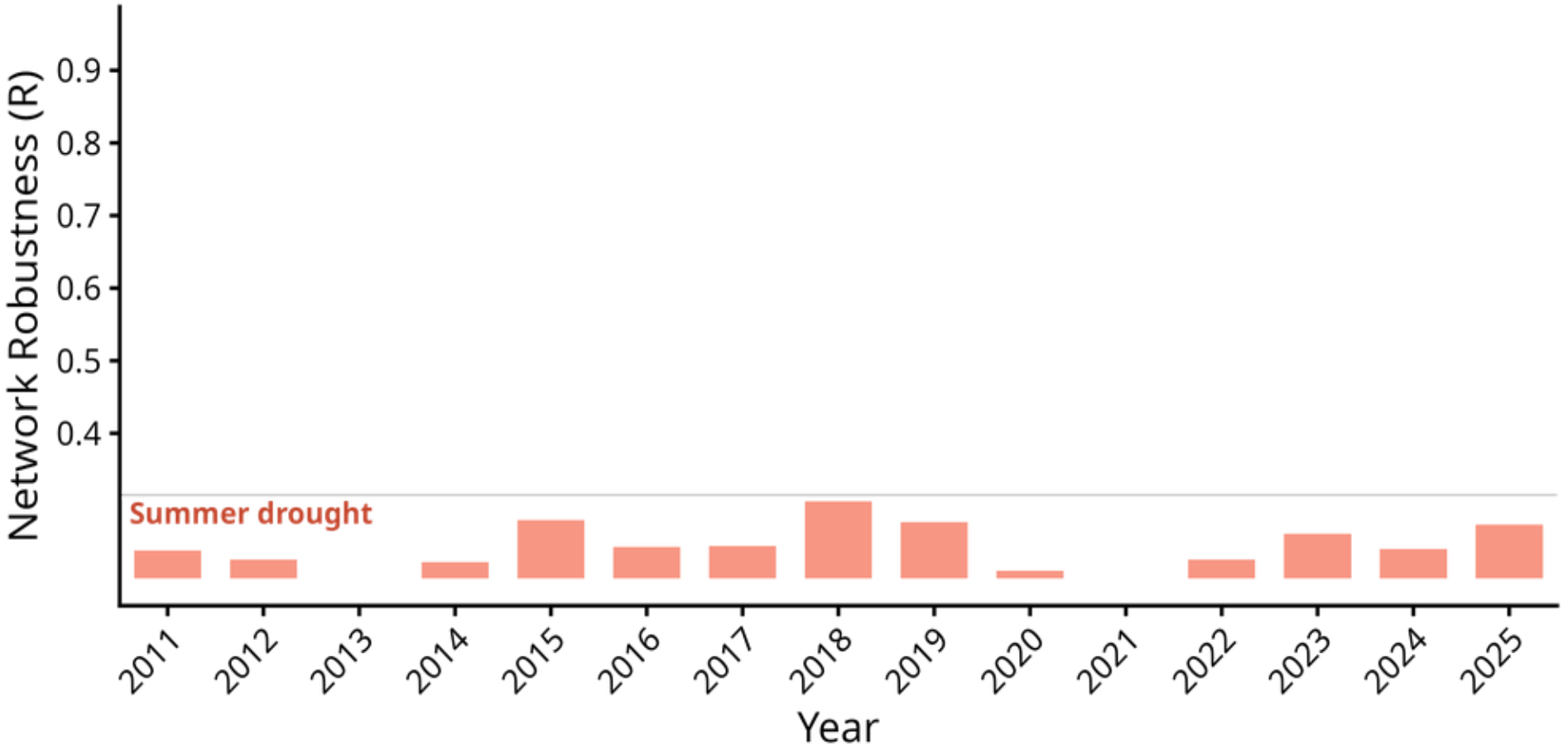


Centrality Variance Explained (Linear Mixed Model)

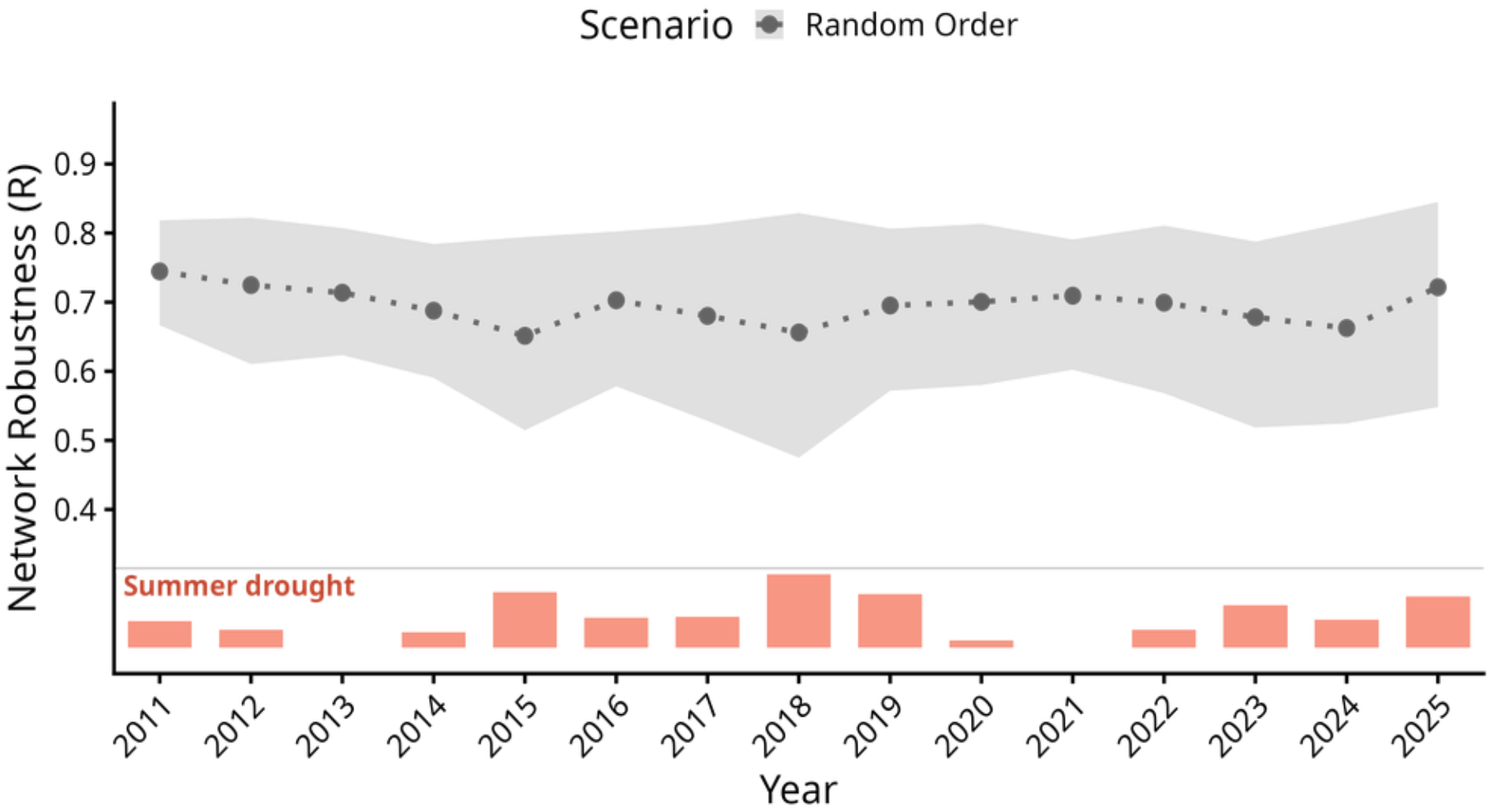
44%
Floral Abundance

38%
Plant Identity ($p = 1.8 \times 10^{-73}$)

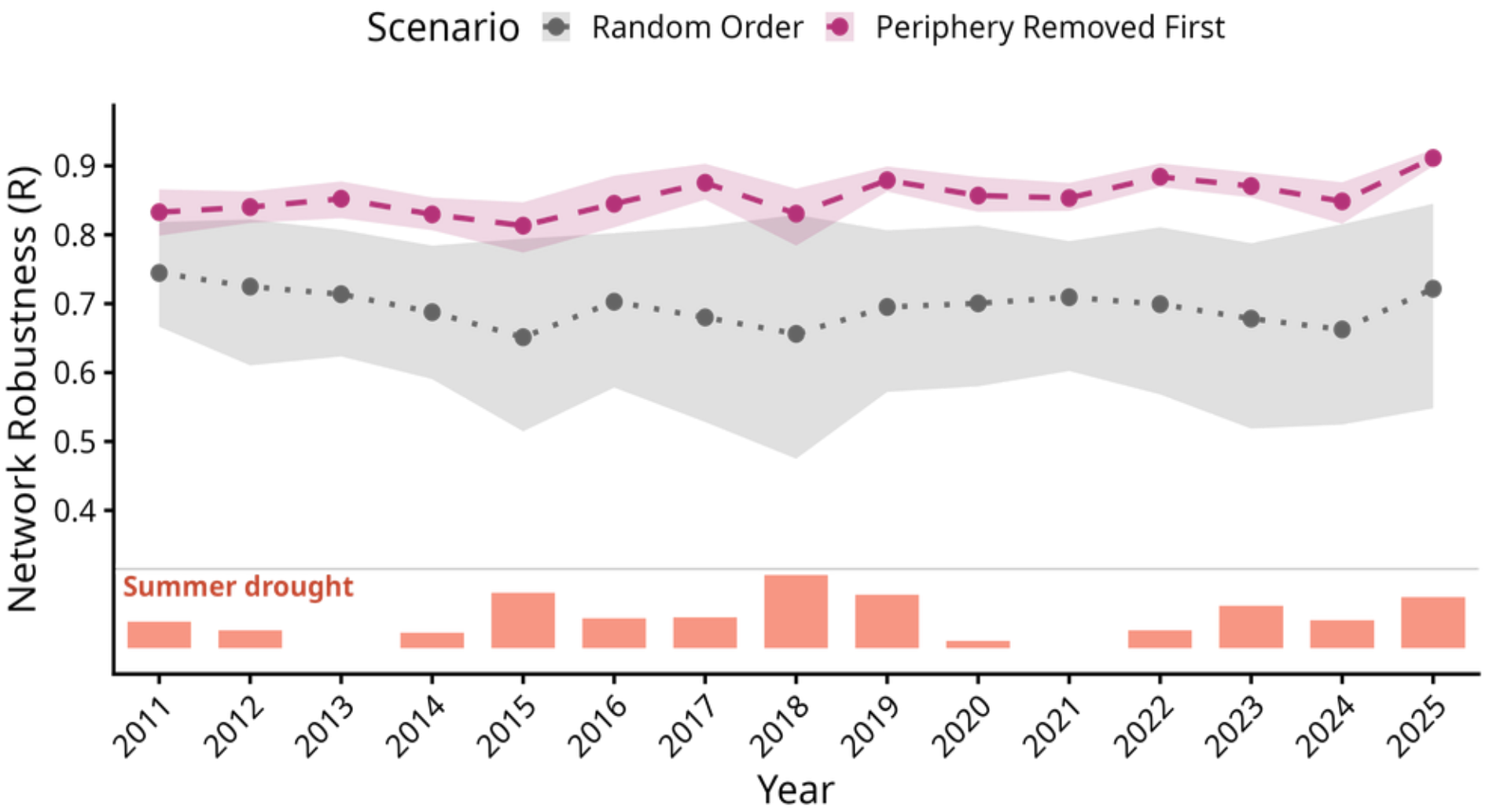
Robustness over Time



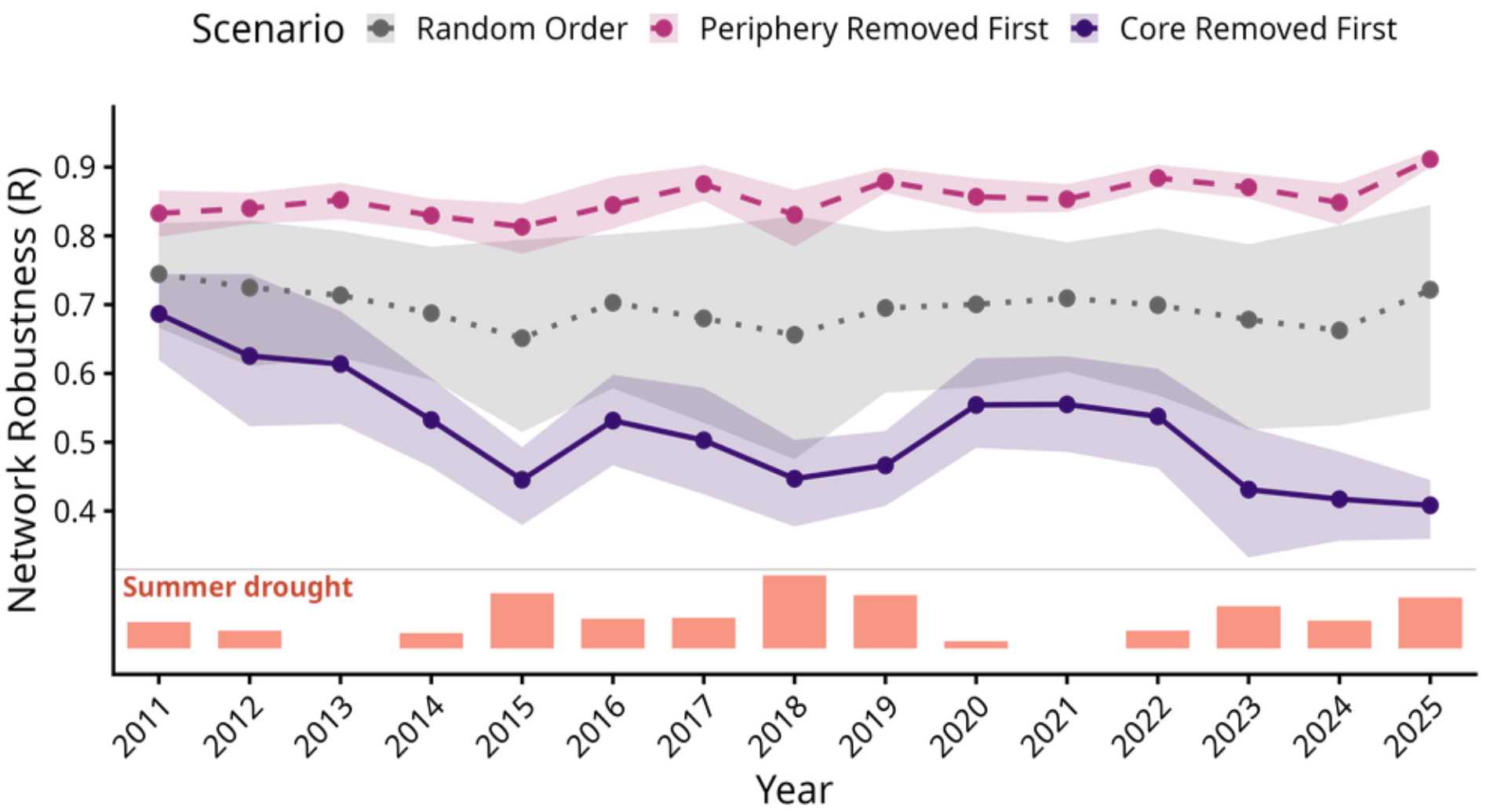
Robustness over Time



Robustness over Time

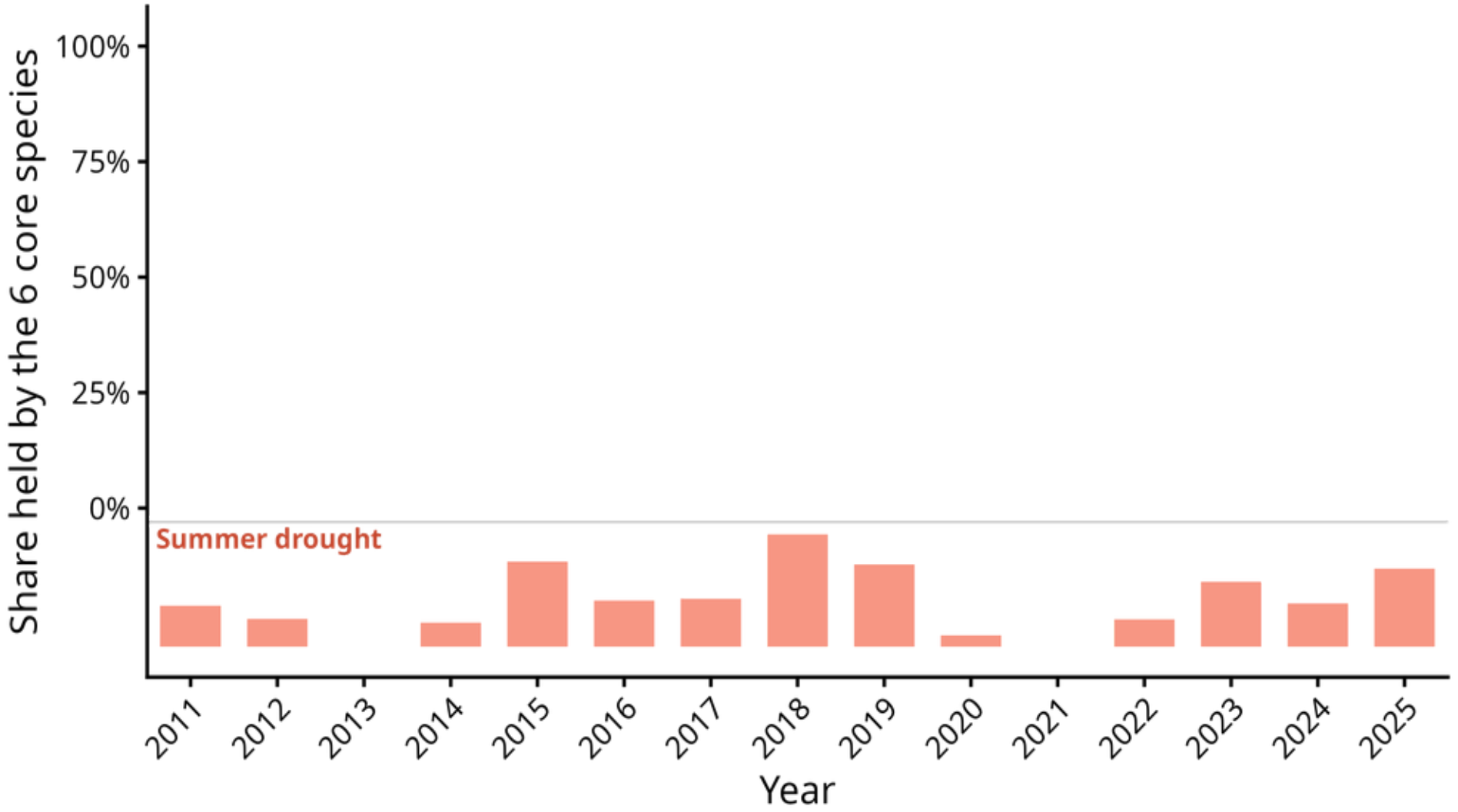


Robustness over Time

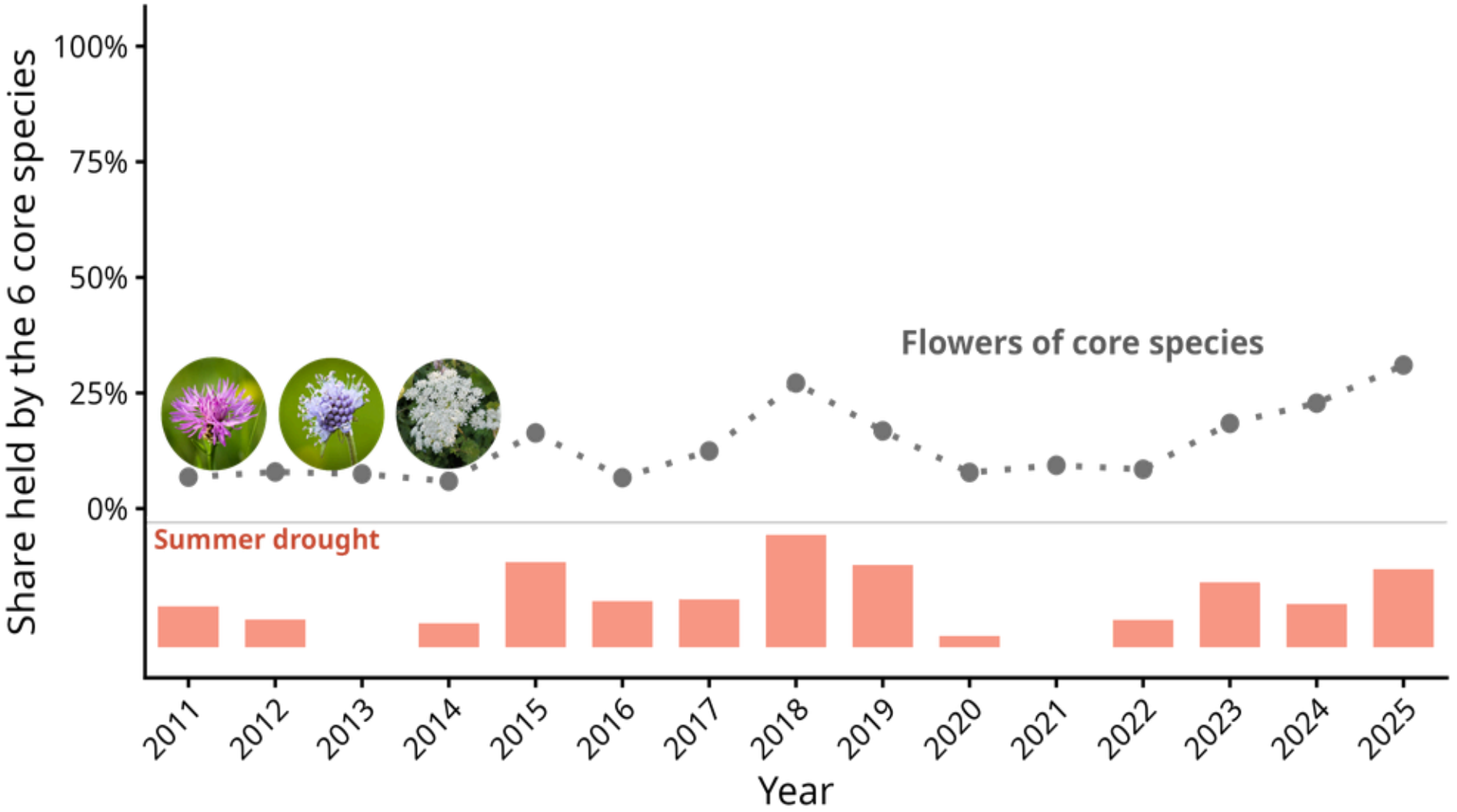


Robustness over Time

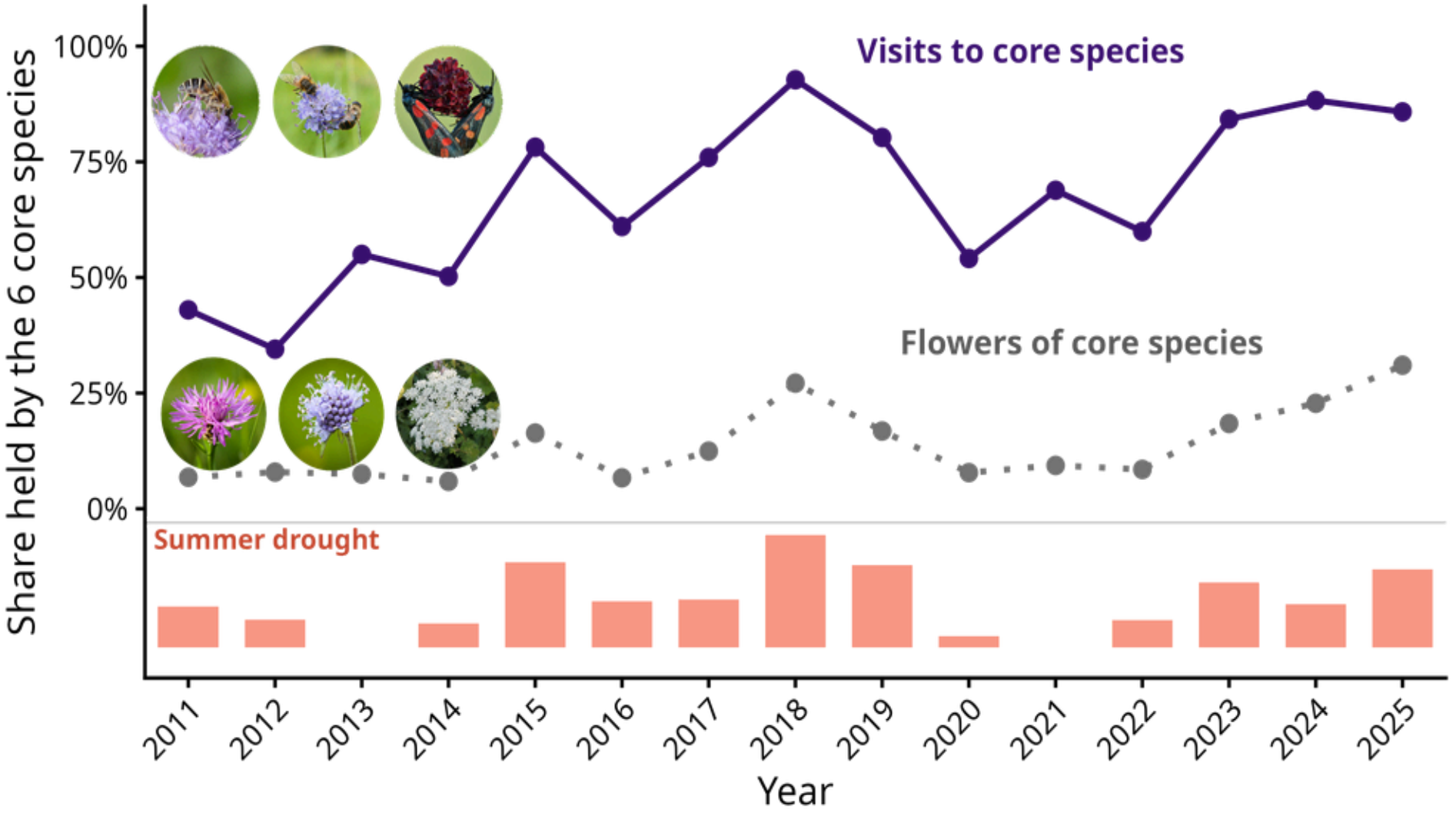
Why?


Share held by the 6 core species
100%
75%
50%
25%
0%
Summer drought
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year

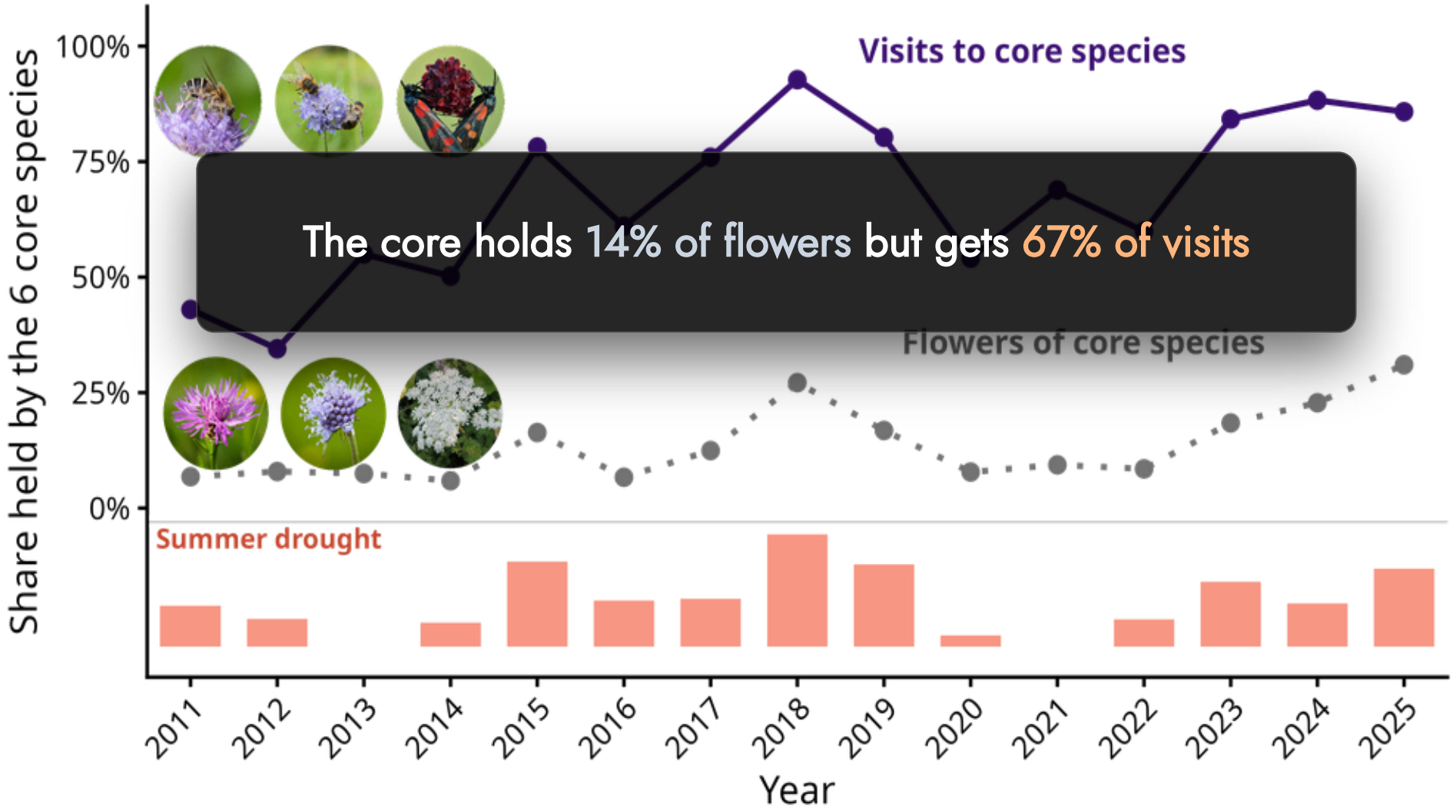
Why?


Share held by the 6 core species
100%
75%
50%
25%
0%
Flowers of core species
Summer drought
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year

Why?

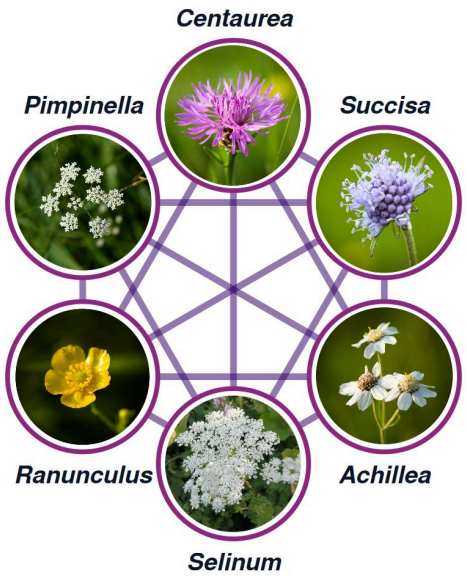

Share held by the 6 core species
100%
75%
50%
25%
0%
Visits to core species
Flowers of core species
Summer drought
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year

Why?


Share held by the 6 core species
100%
75%
50%
25%
0%
Visits to core species
The core holds 14% of flowers but gets 67% of visits
Flowers of core species
Summer drought
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
Year

Take-Home Messages

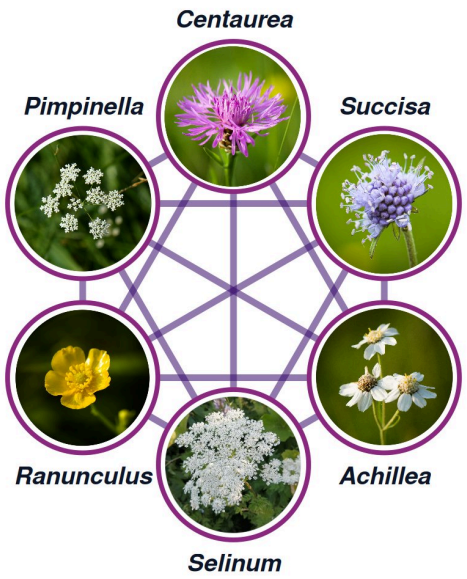
01 Persistent Core



6 species stay central for 15 years

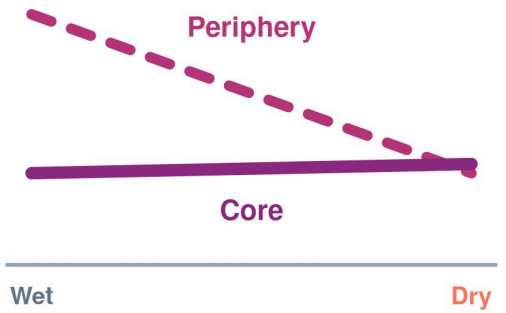
Take-Home Messages

01 Persistent Core



6 species stay central for 15 years

02 Rising Dependence



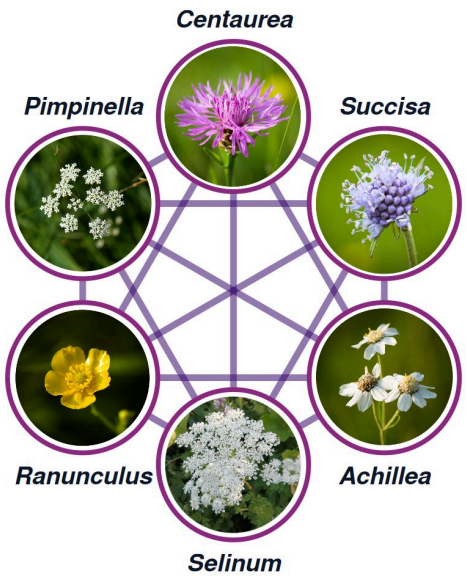
46% → 86%

of visits go to the core

Drought shifts pollinators onto the core

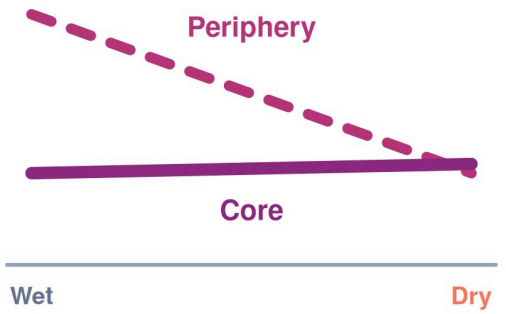
Take-Home Messages

01 Persistent Core



6 species stay central for 15 years

02 Rising Dependence



46% → 86%

of visits go to the core

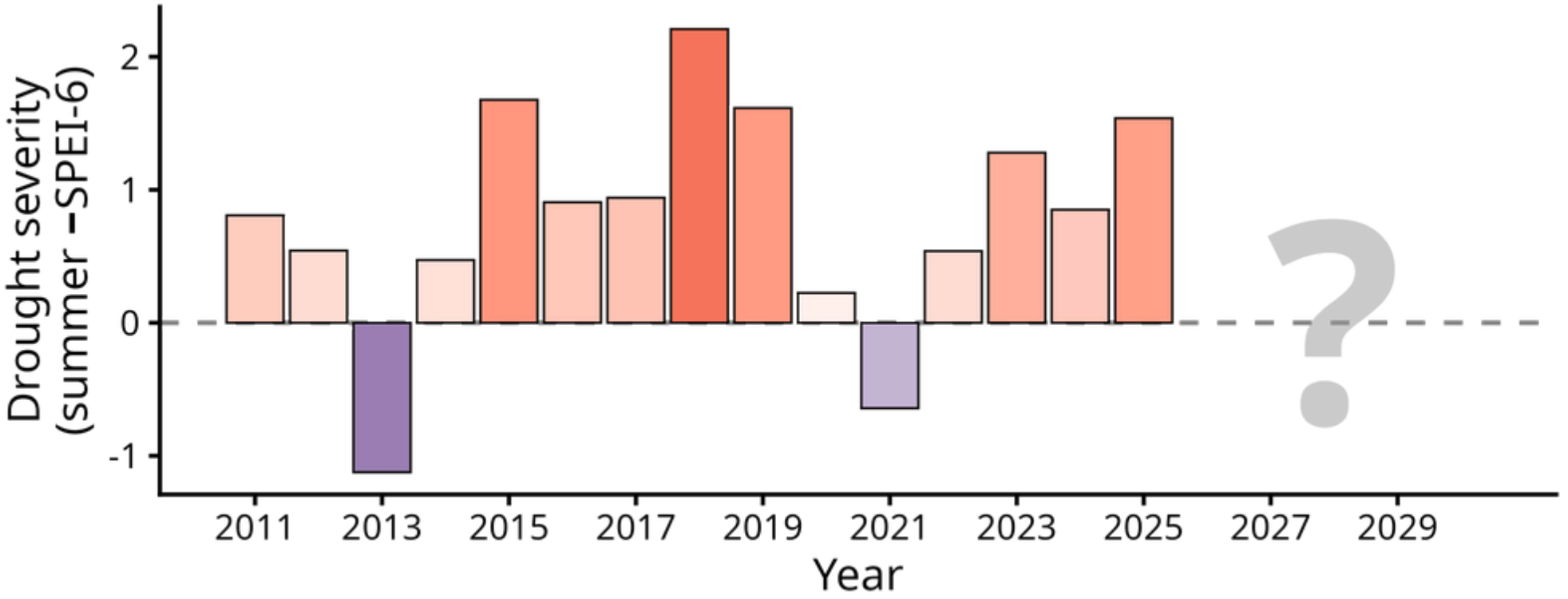
Drought shifts pollinators onto the core

03 Protect the Core

Should we protect species that stay central and keep flowering through perturbations?

Future Steps?

Future Steps?



Data: TerraClimate (Abatzoglou et al., 2018), Handrkov locality (2011–2025)

Next Step: Pollen Transfer

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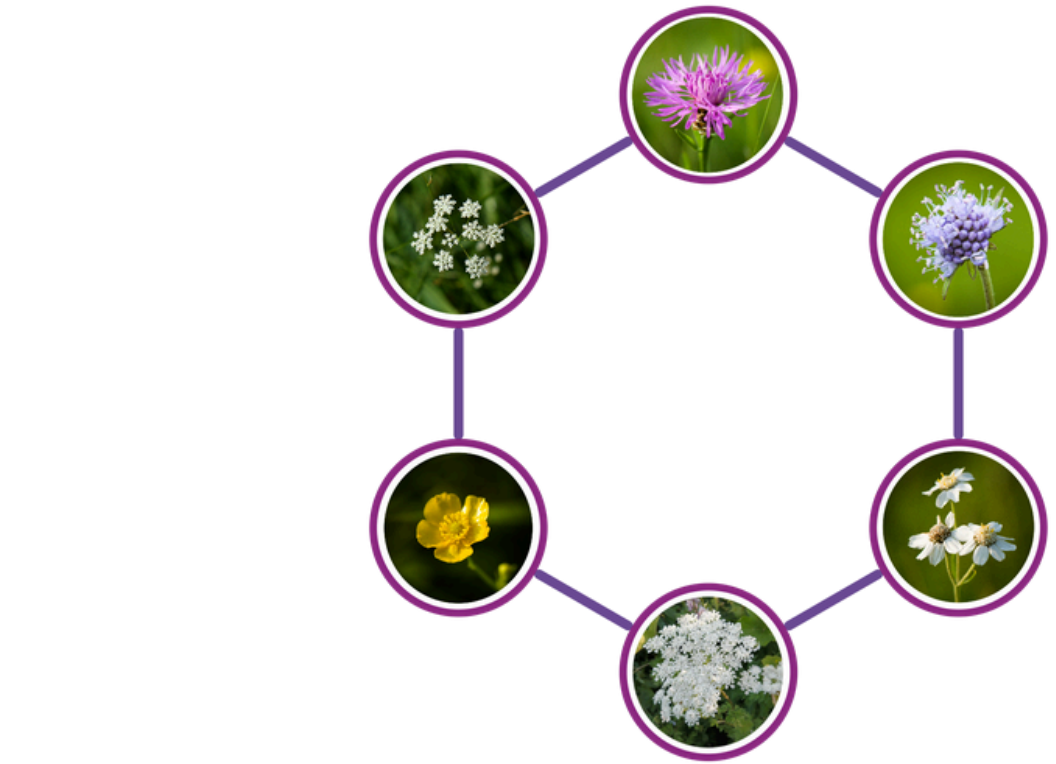
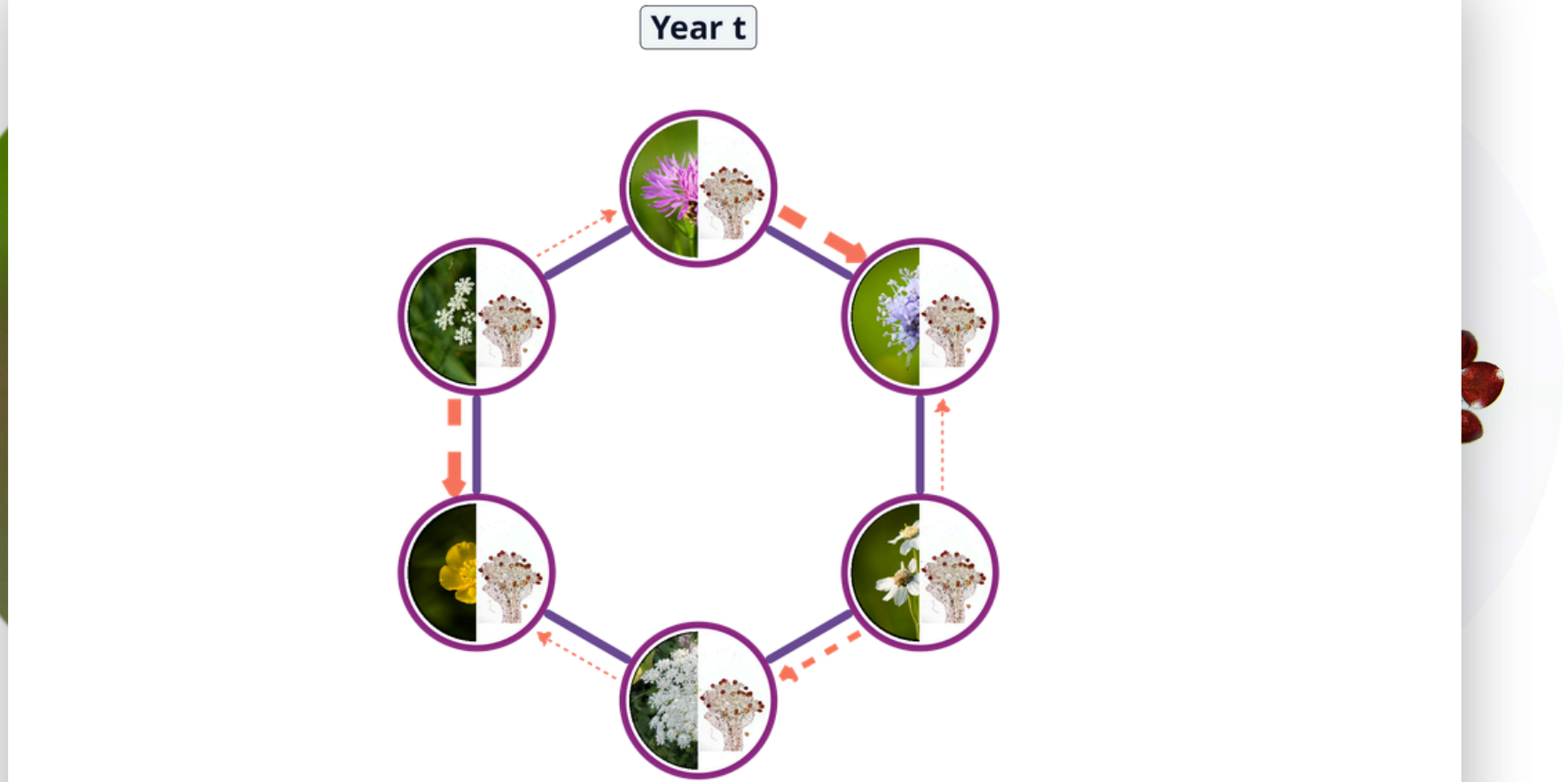
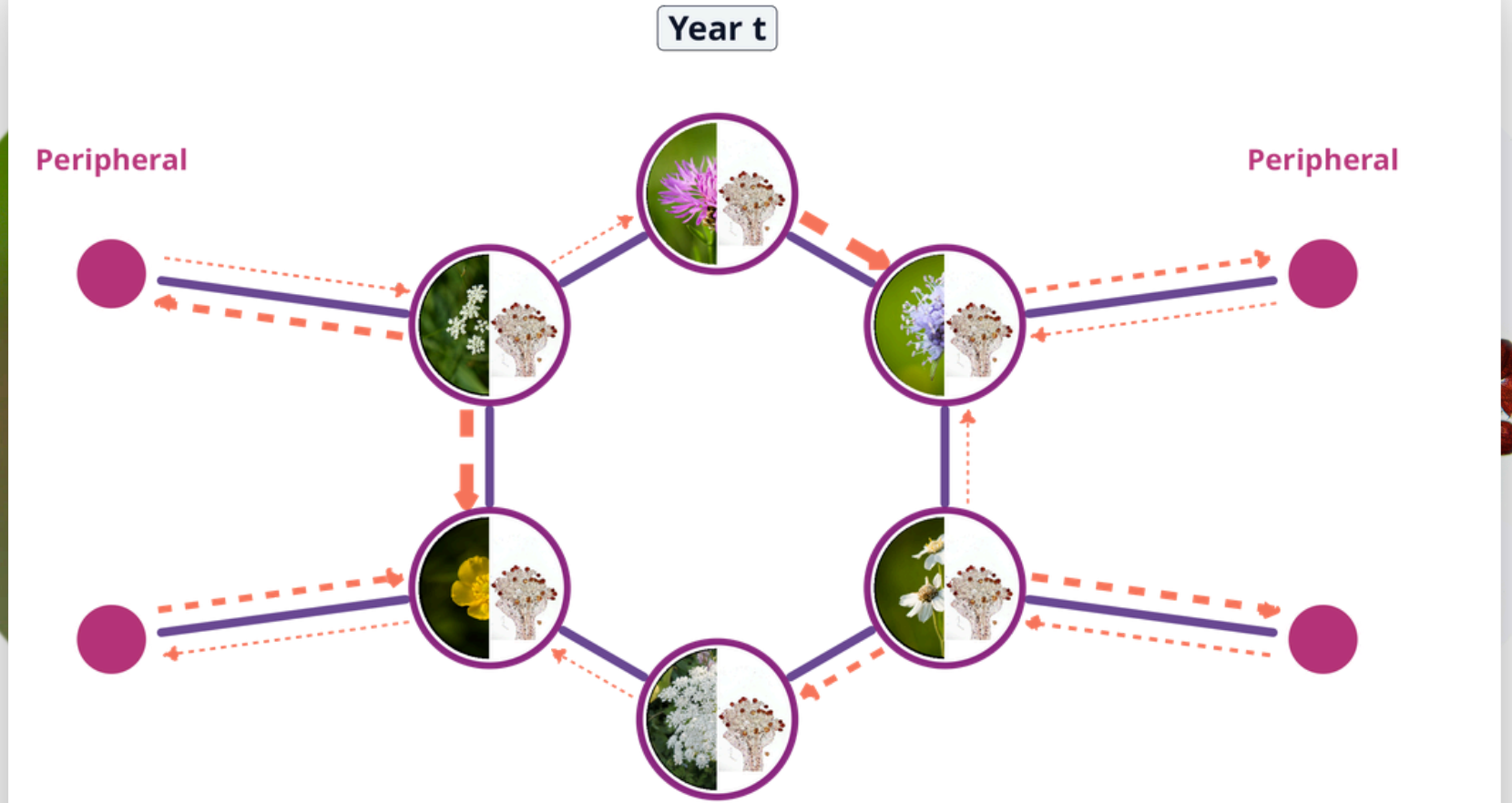


shared visitors

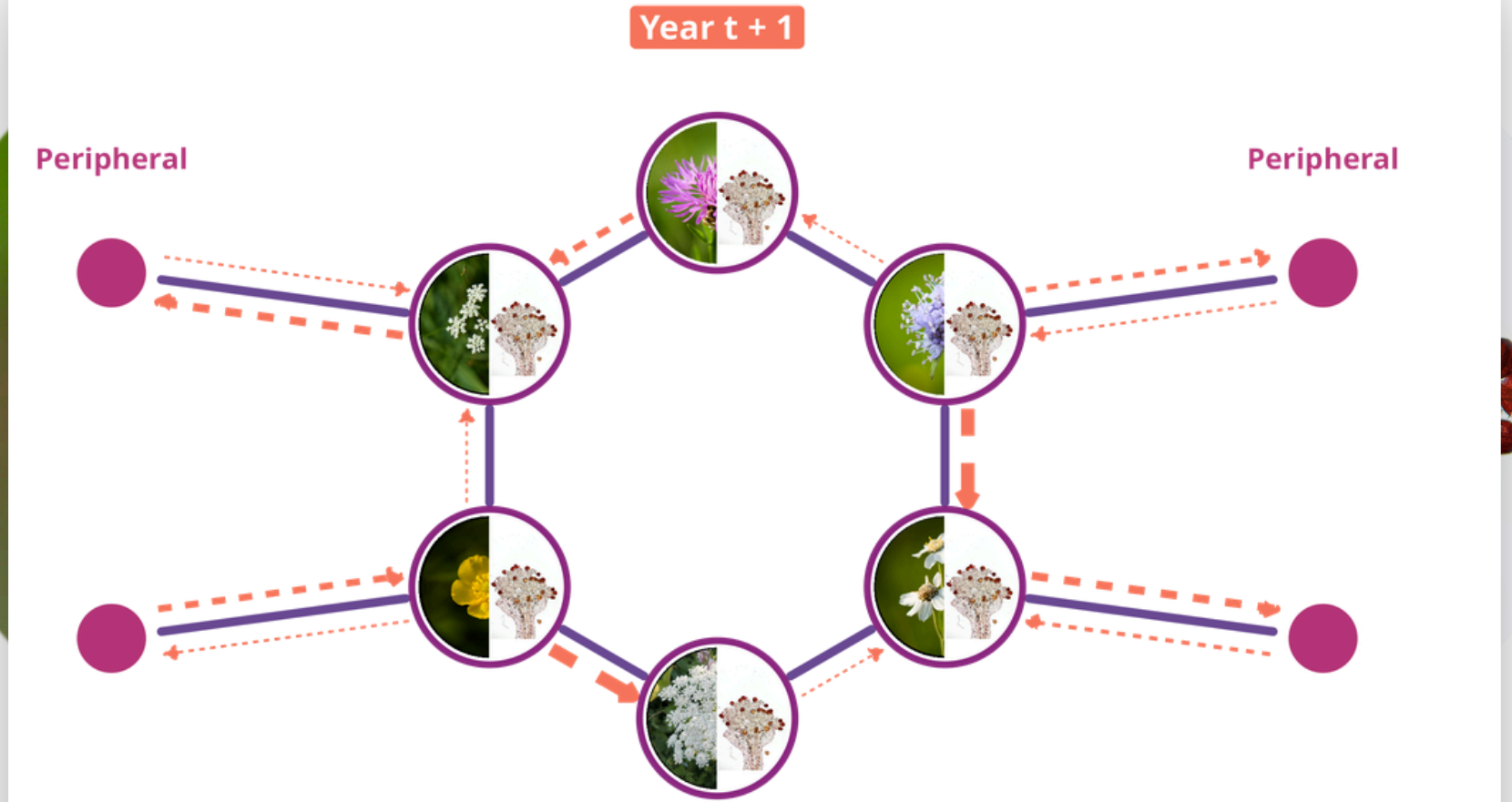
Next Step: Pollen Transfer



Next Step: Pollen Transfer



Next Step: Pollen Transfer



Handrkov

Thanks to over 100 people!

Handrkov - Collaboration

Handrkov - Collaboration

Handrkov - Collaboration

Handrkov - Collaboration

Ecoflor 2027!

Ecoflor 2027!

Register now!

pollination-ecology-group.github.io/ECOFLOR2027/

Thank you for your attention!

Jakub Štenc

jakubstenc@gmail.com | +420 774 553 734

Charles University, Prague & CREAF, Barcelona

Save the date: Ecoflor 2027 in Prague!

Ecoflor 2027

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Appendix & Backup Slides

Q&A: Is It Circular?

The Circularity Skeptic: *"Core species are high-degree hubs by definition. Doesn't removing them automatically reduce robustness mathematically, making core importance a tautology?"*

PILLAR 1

Climatic Decoupling

If circular, core removal would cost the same every year.

Instead, core-loss robustness ranges from **0.69 to 0.41** and drops in **drier summers** ($p = 0.005$).

Weather cannot change a mathematical tautology.

PILLAR 2

Static vs. Dynamic Gap

Core identity is **static** (the same 6 species across 15 yrs from null models).

Yet the fragility gap (ΔR) is **dynamic**: it surges with summer drought ($p = 0.018$) and intensifies over 15 years ($p = 0.014$).

A static definition cannot generate a 15-year temporal trend.

PILLAR 3

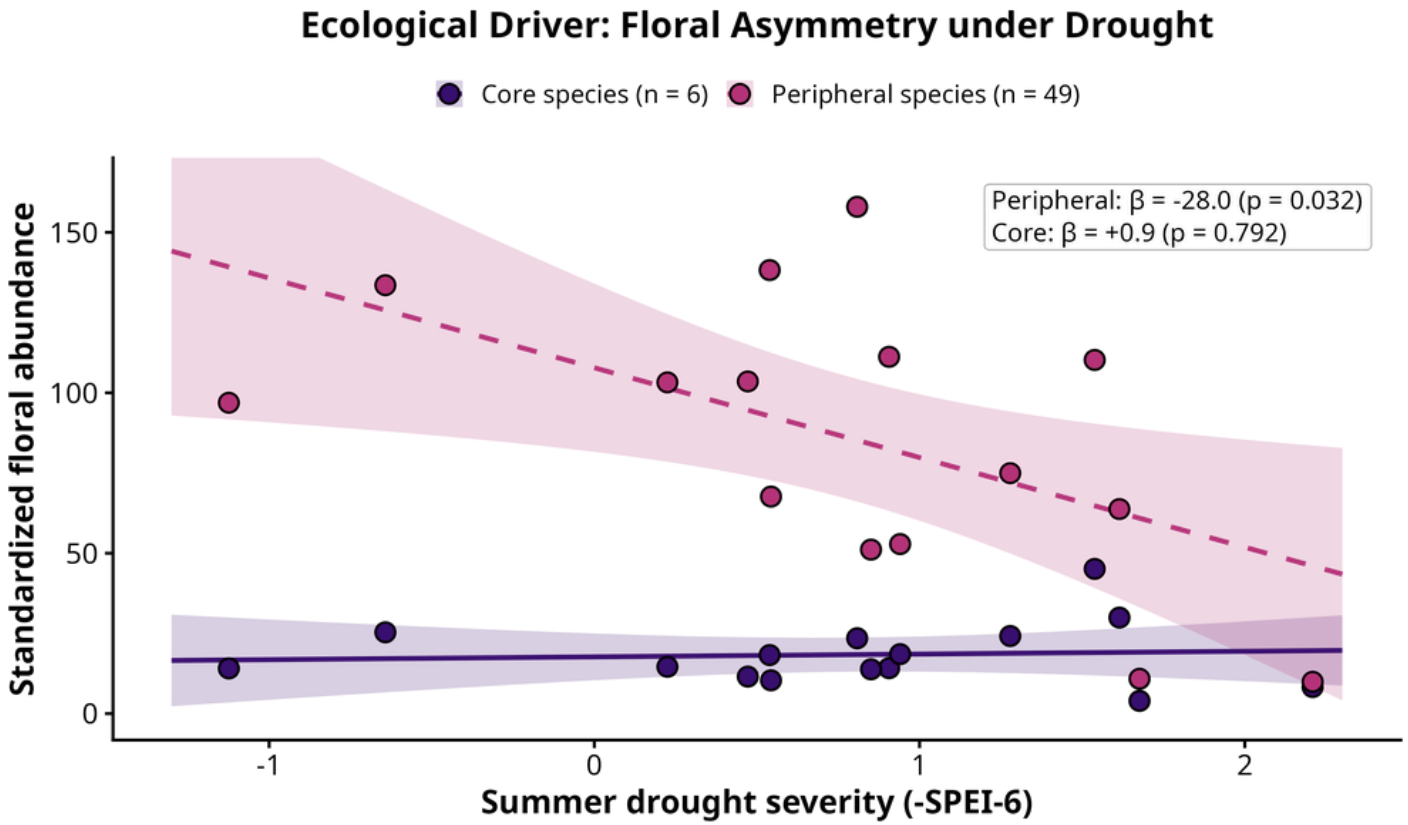
Empirical Biology

Driven by **plant physiology**, not network geometry:

Peripheral flowering drops **~3×** along the drought gradient ($\beta = -28.0$, $p = 0.032$), while core flowering does not.

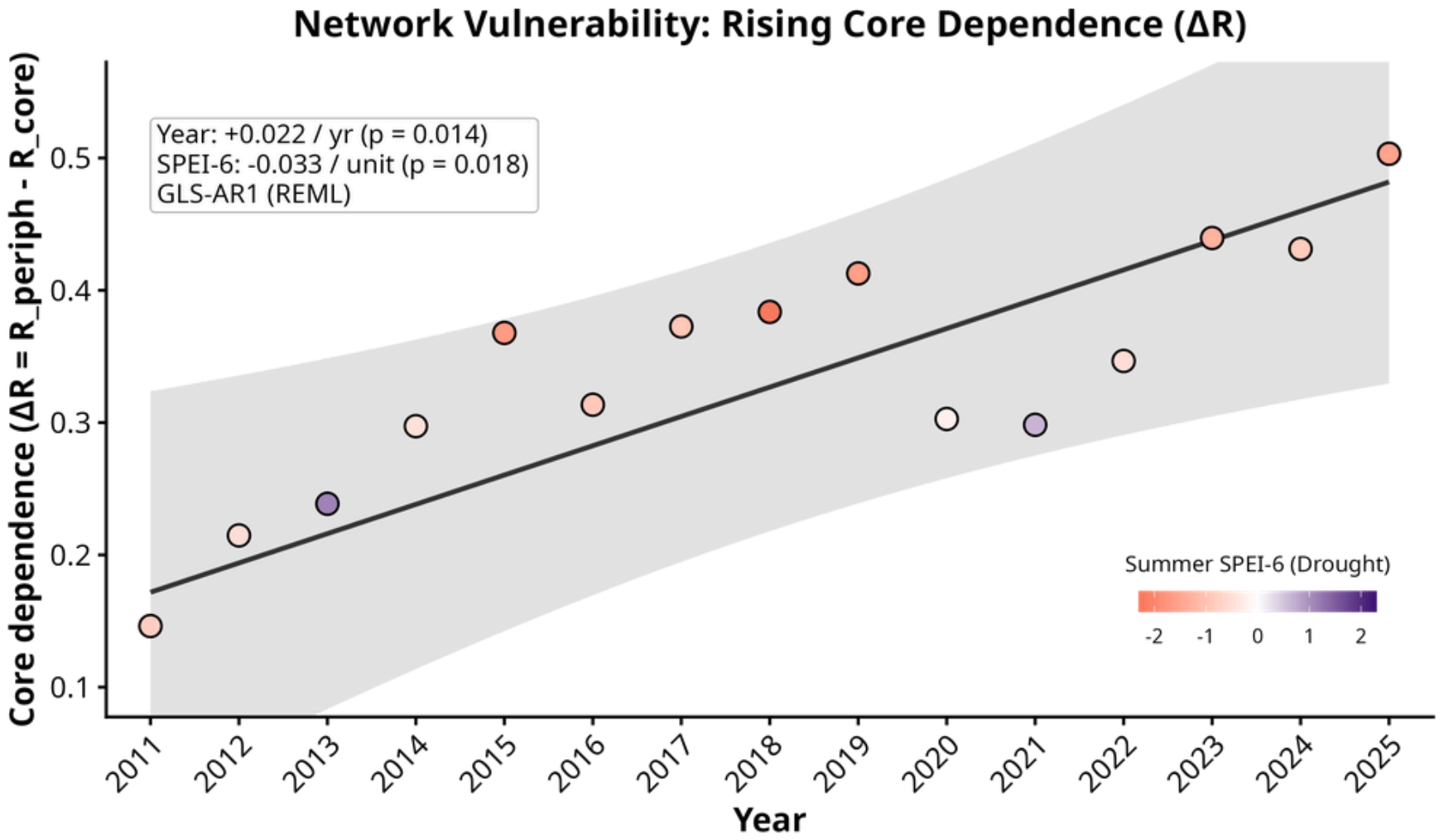
Pollinators concentrate onto the core (34–55% → 84–88% of visits), so losing it removes most of the remaining forage.

Appendix: Floral Response to Drought

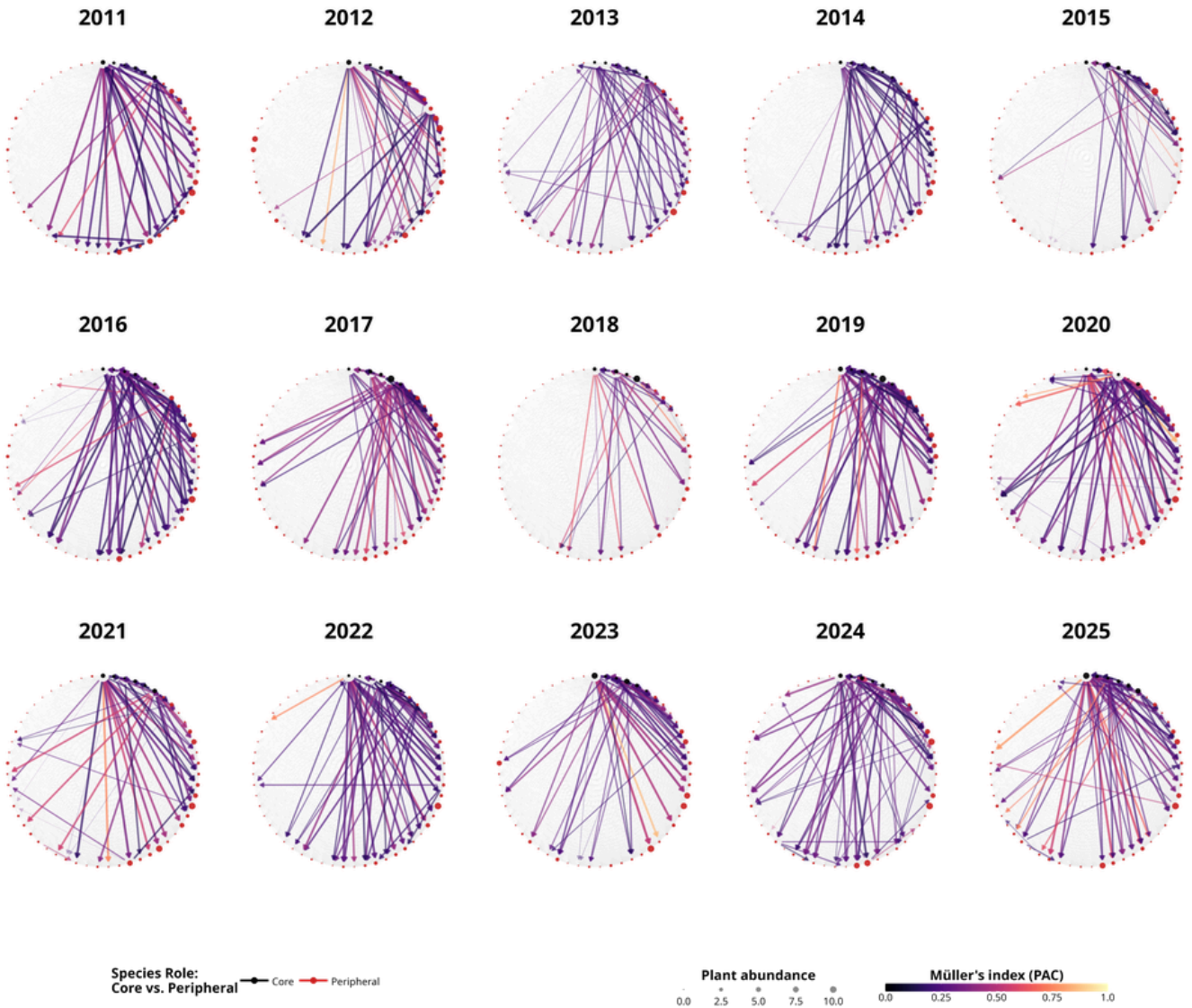


The Life-Raft Mechanism: Peripheral flowering drops ~3× across the drought gradient (β = −28.0, p = 0.032), core flowering shows no detectable drop (β = +0.9, p = 0.79) → pollinators concentrate on the core: from 34–55% of visits (2011–14) to 84–88% (2023–25).

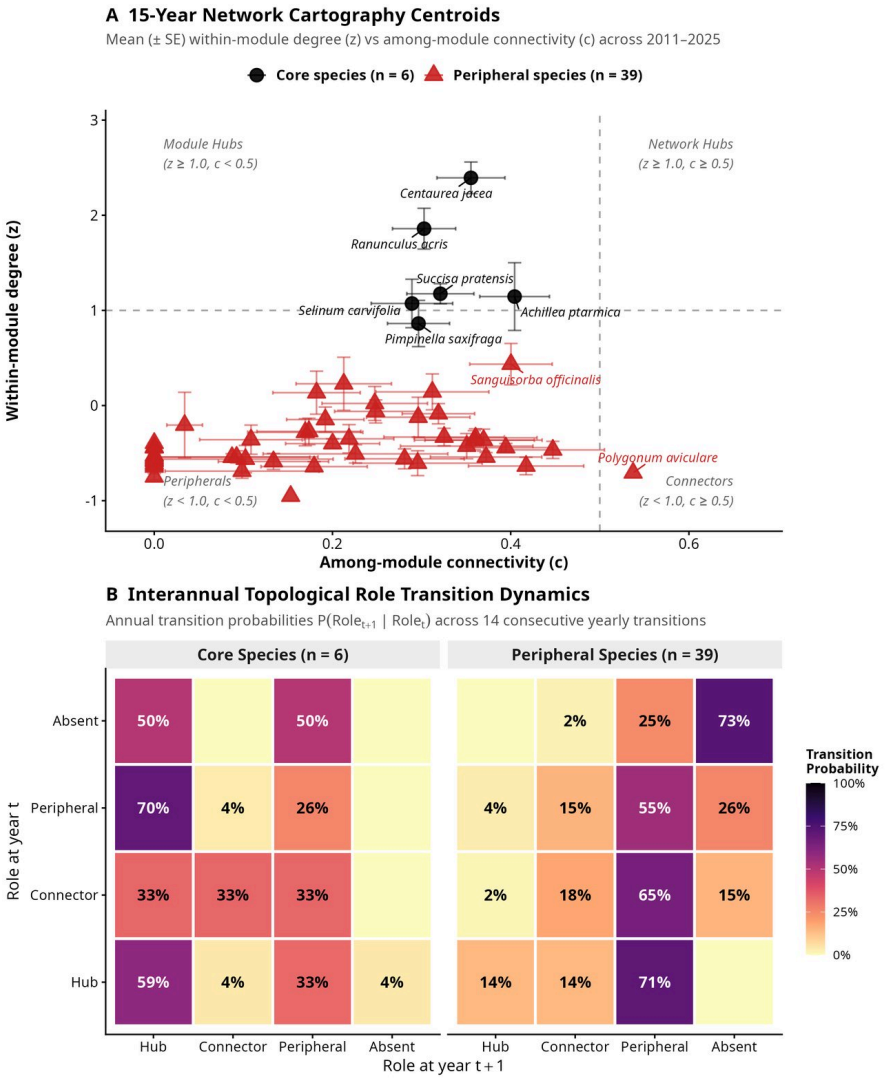
Appendix: Core Dependence (ΔR)



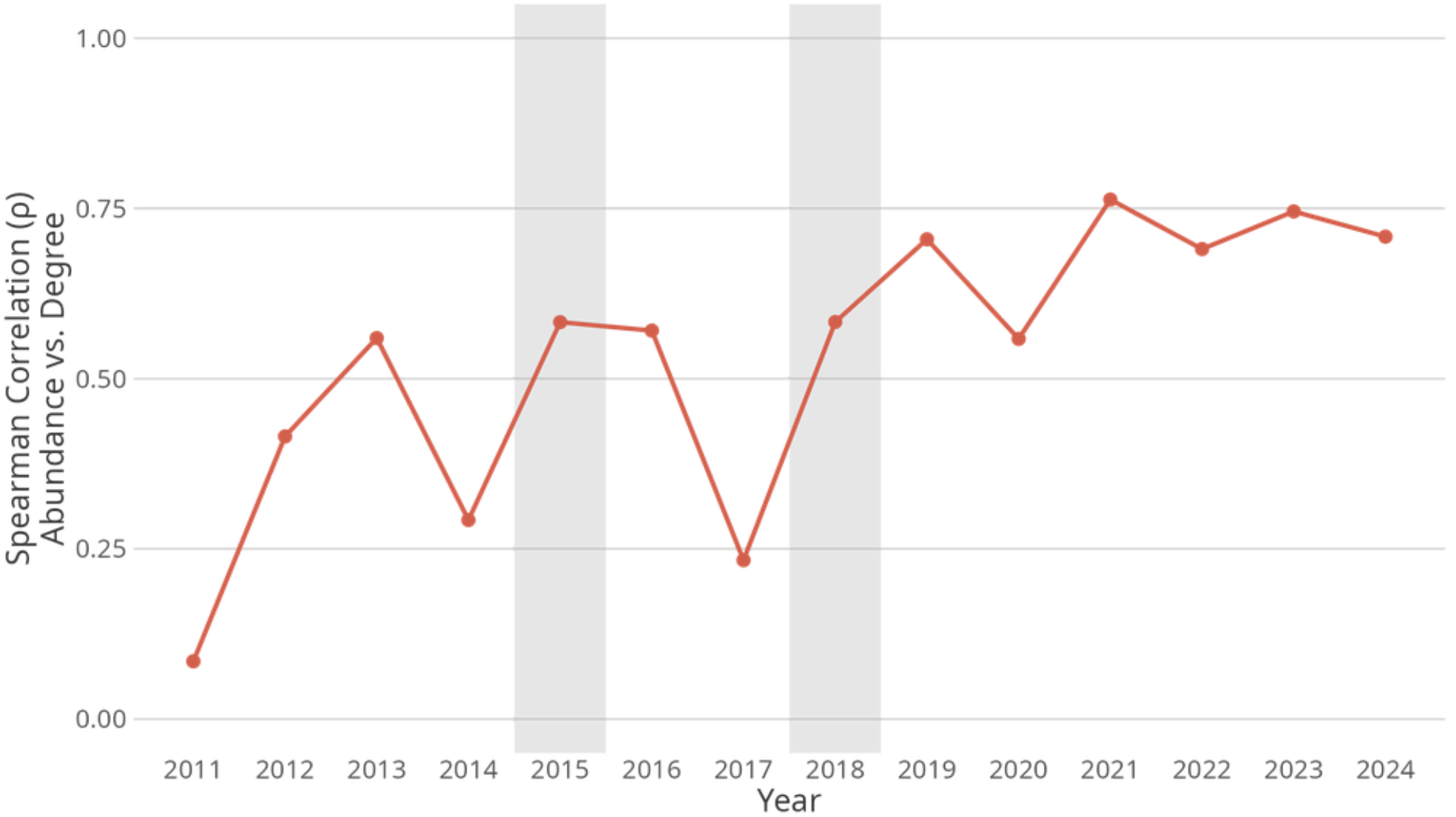
Appendix: 15-Year Network Snapshots



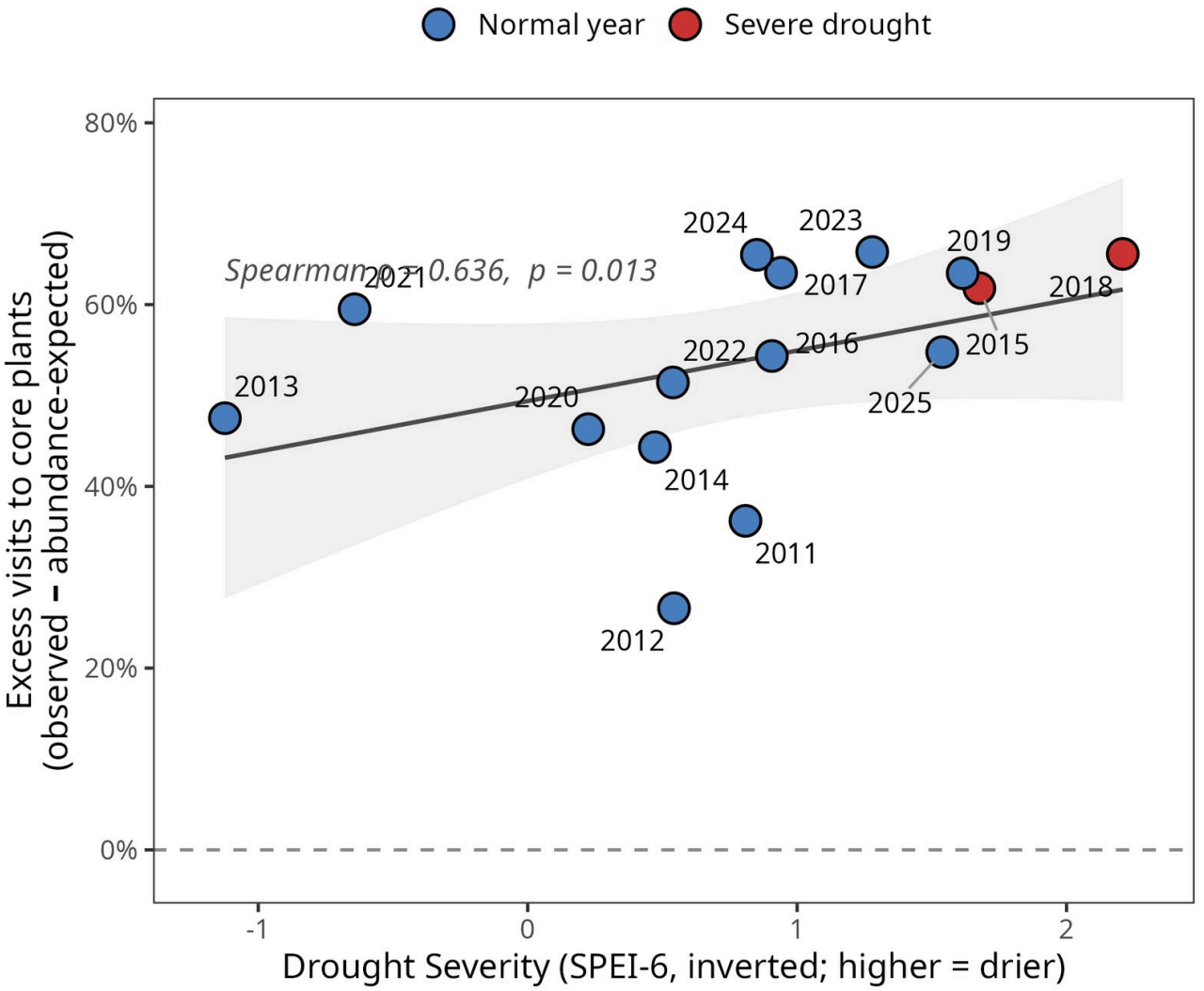
Appendix: Species Roles



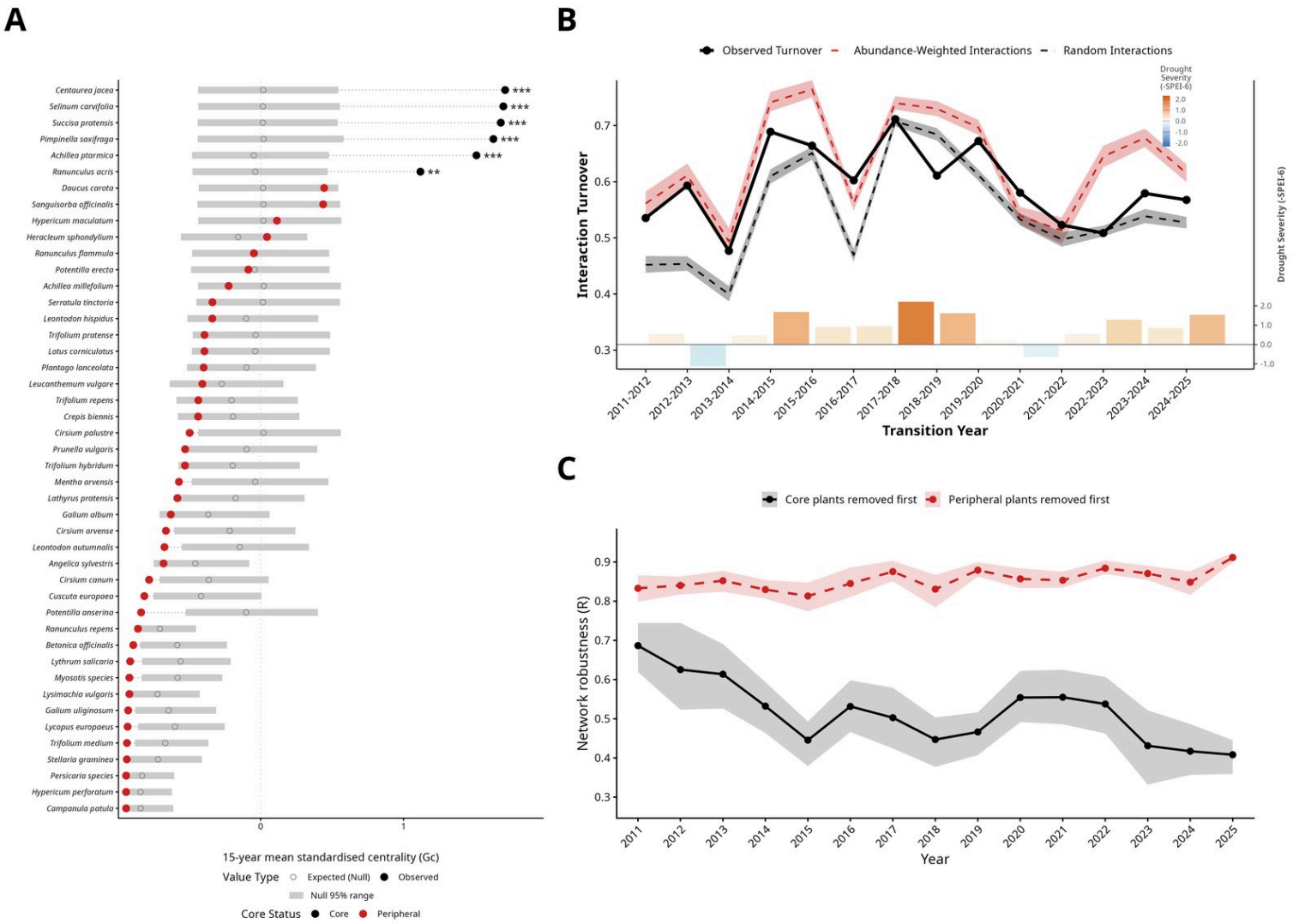
Appendix: Abundance vs Degree Correlation



Appendix: Pollinator Concentration



Appendix: Turnover vs Null Models



Appendix: Mechanism (Figure 3)